Lesson 03 Demo 03

Creating and Scaling EKS Clusters

Objective: To create and scale an AWS EKS cluster, configuring it for secure, efficient, and scalable deployment of containerized applications

Tools required: AWS Cloud and EKS Service

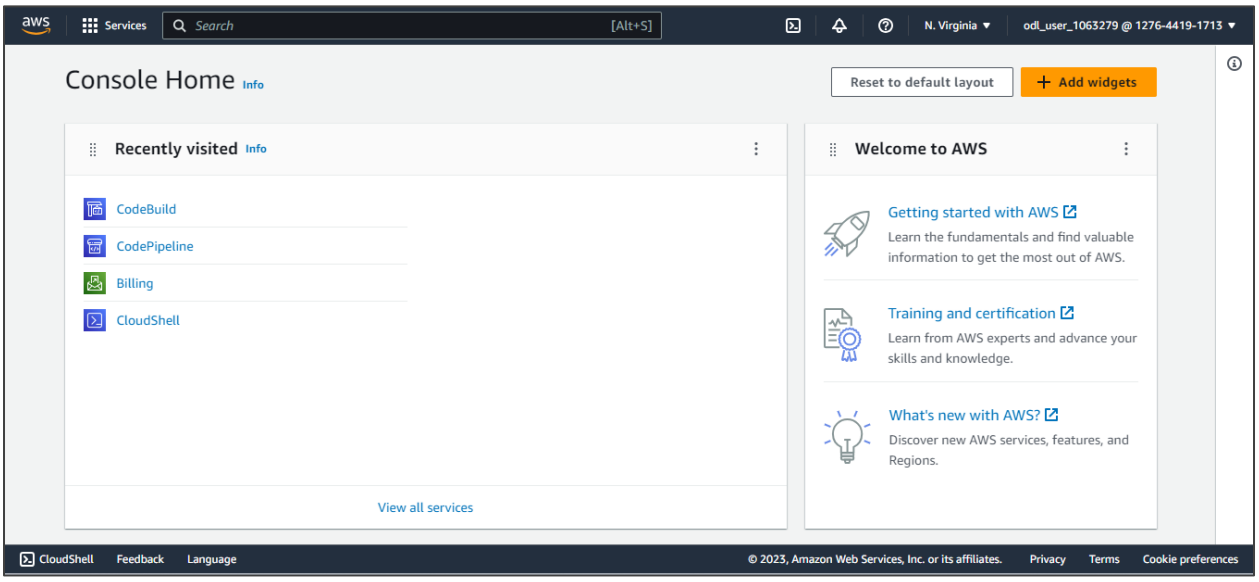
Prerequisites: None

Steps to be followed:

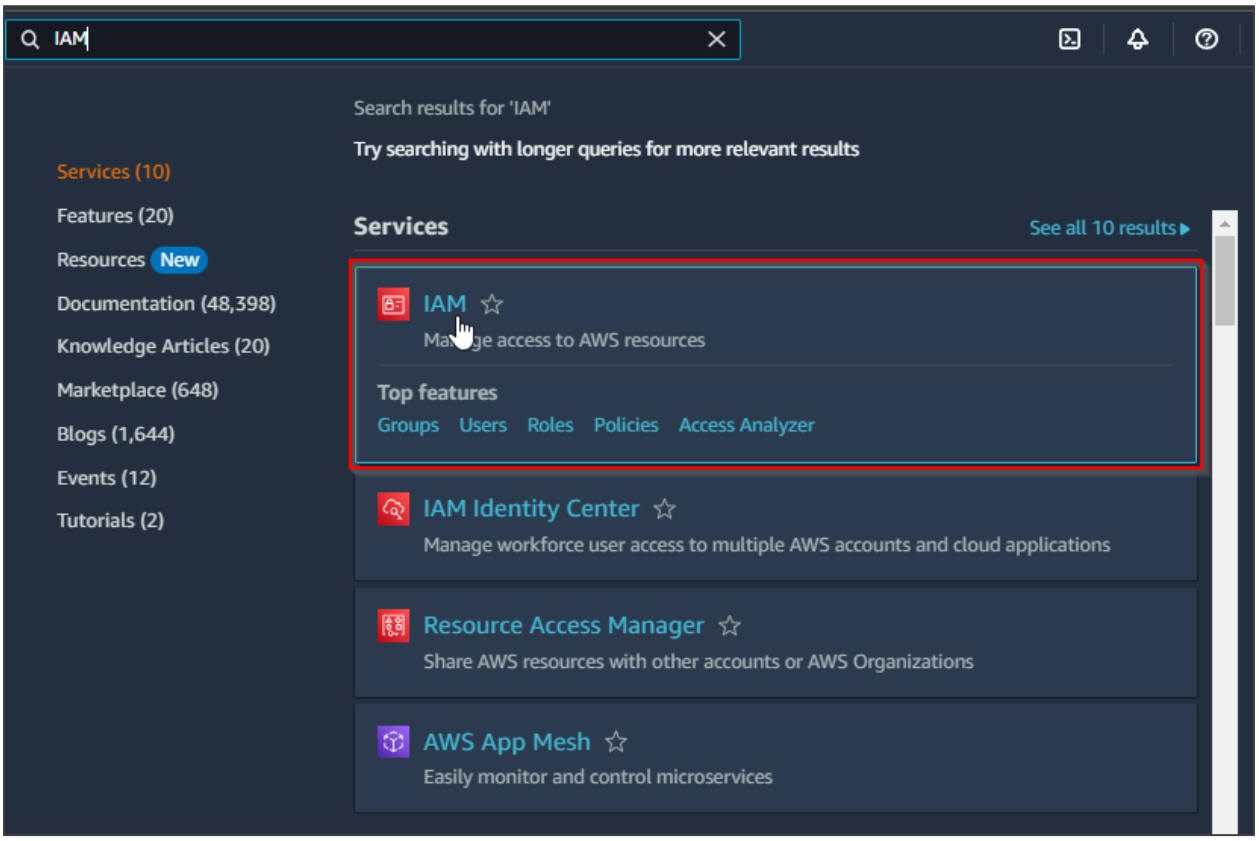
1. Create the IAM role for the Amazon EKS cluster and EKS worker node group
2. Create the AWS EKS cluster
3. Create the AWS EKS worker node group

Step 1: Create the IAM role for the Amazon EKS cluster and EKS worker node group

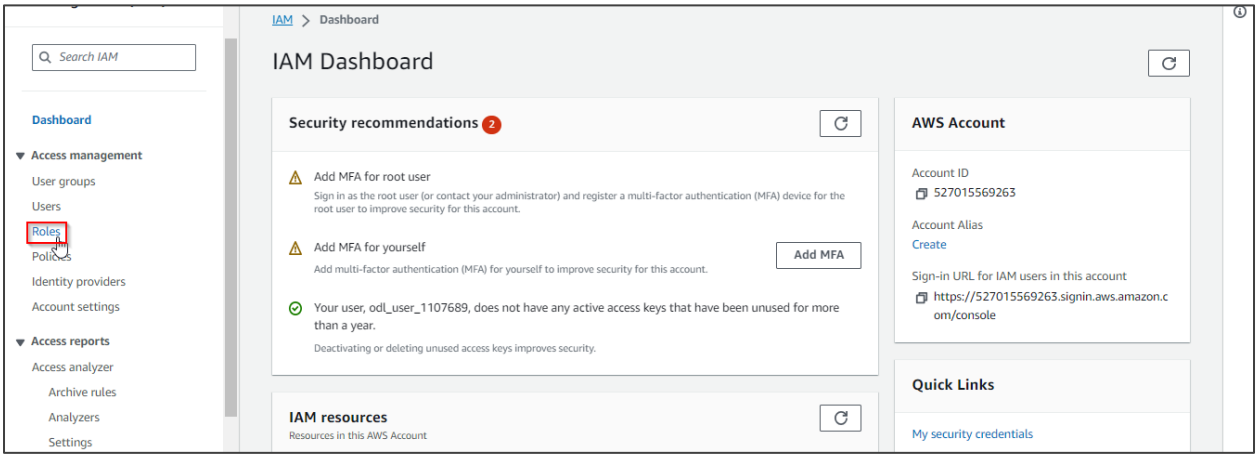
1.1 Sign in to the AWS console



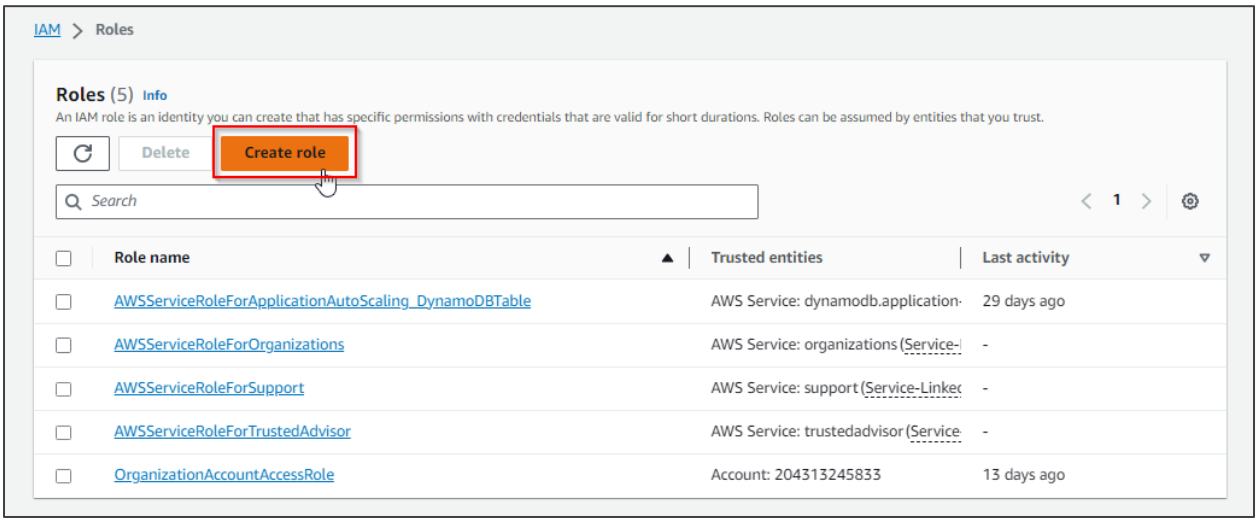
1.2 Search for and click on **IAM**



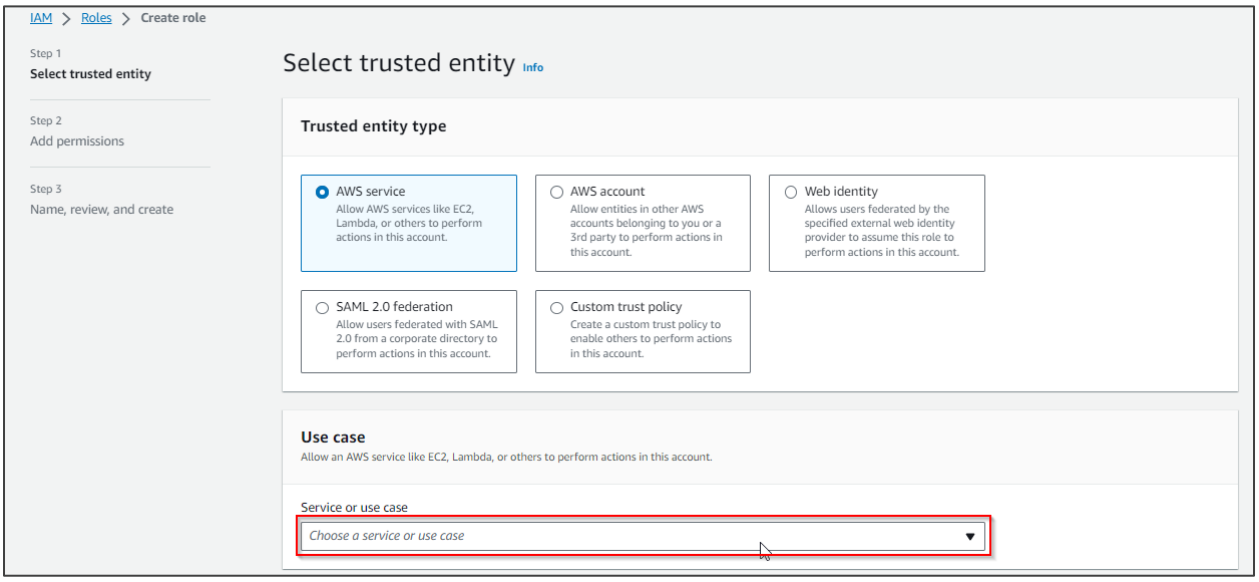
1.3 In the list on the left side, click on **Roles**

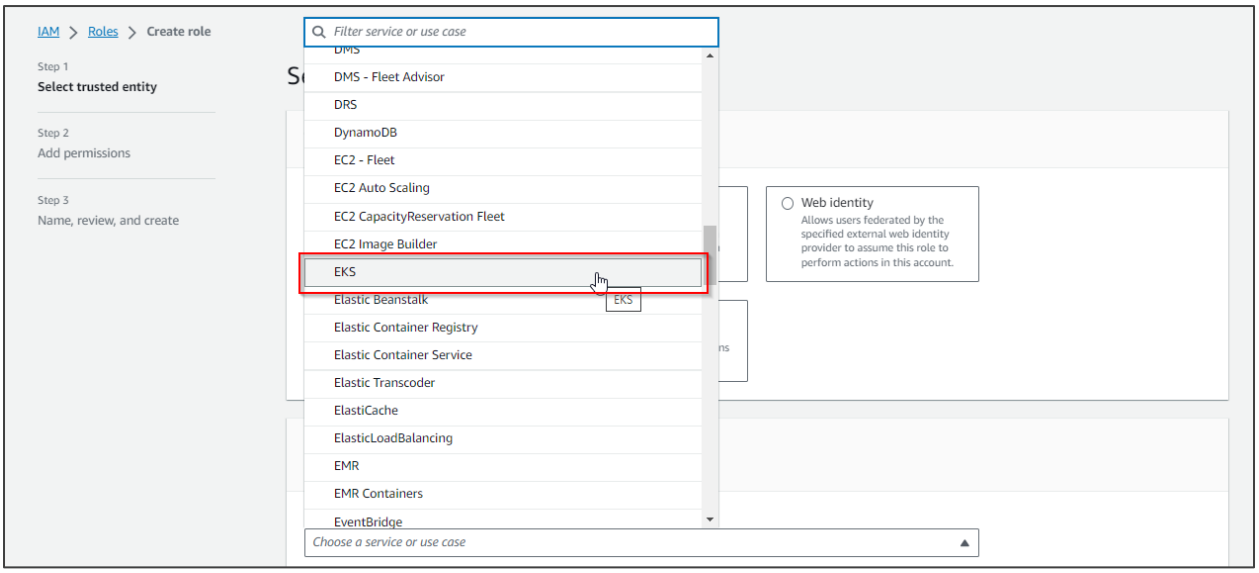


1.4 Click on the **Create role** button

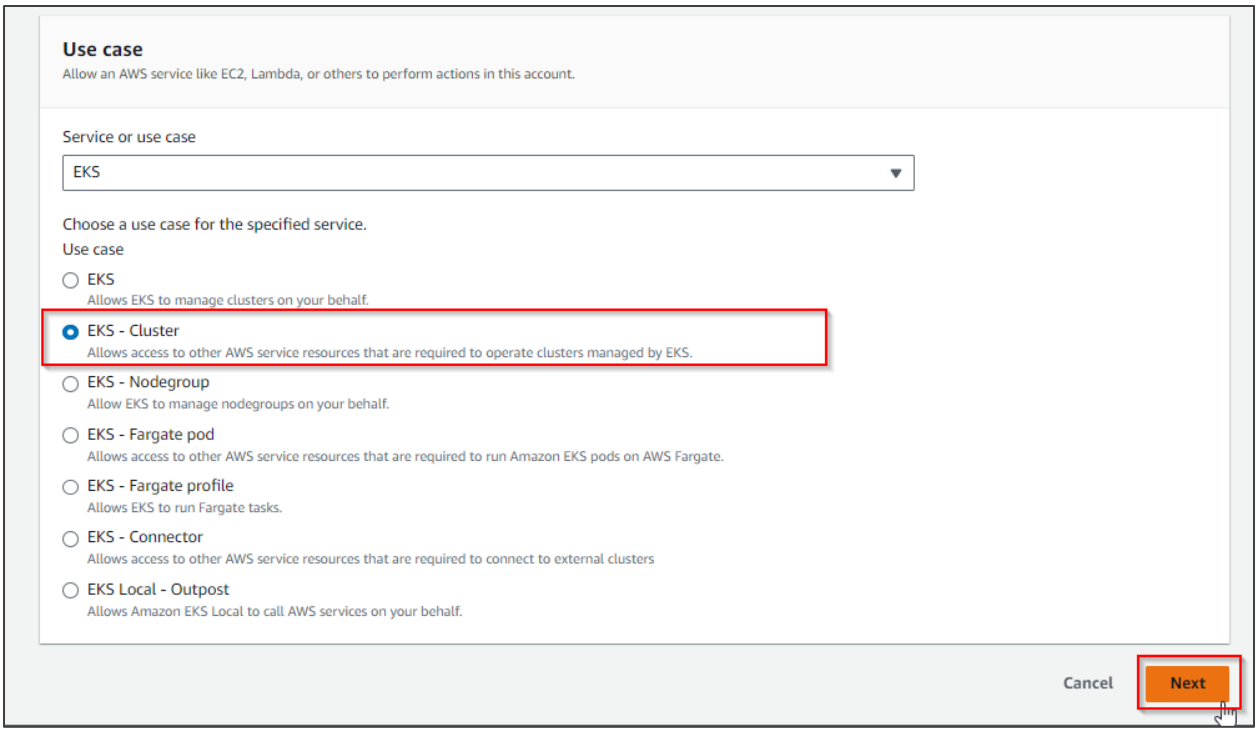


1.5 In the **Use case** section, select **EKS** service from the drop-down list

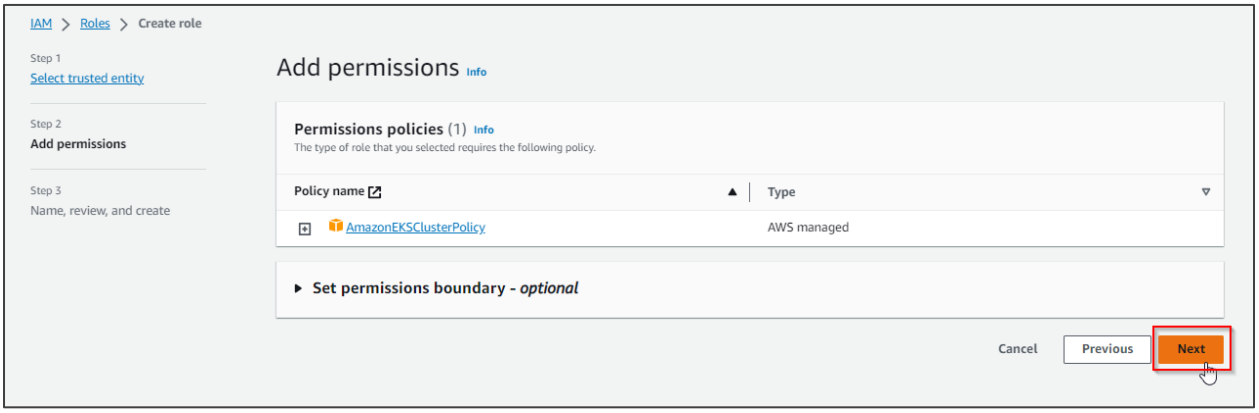




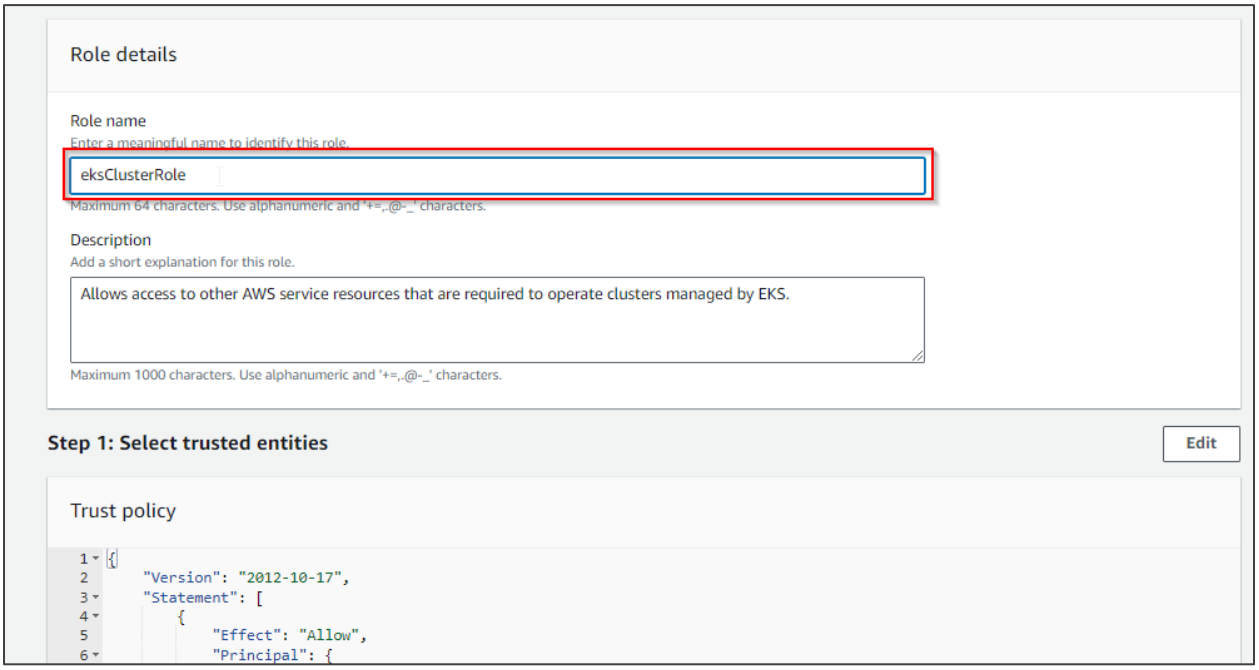
1.6 Now, select **EKS – Cluster** and click on the **Next** button



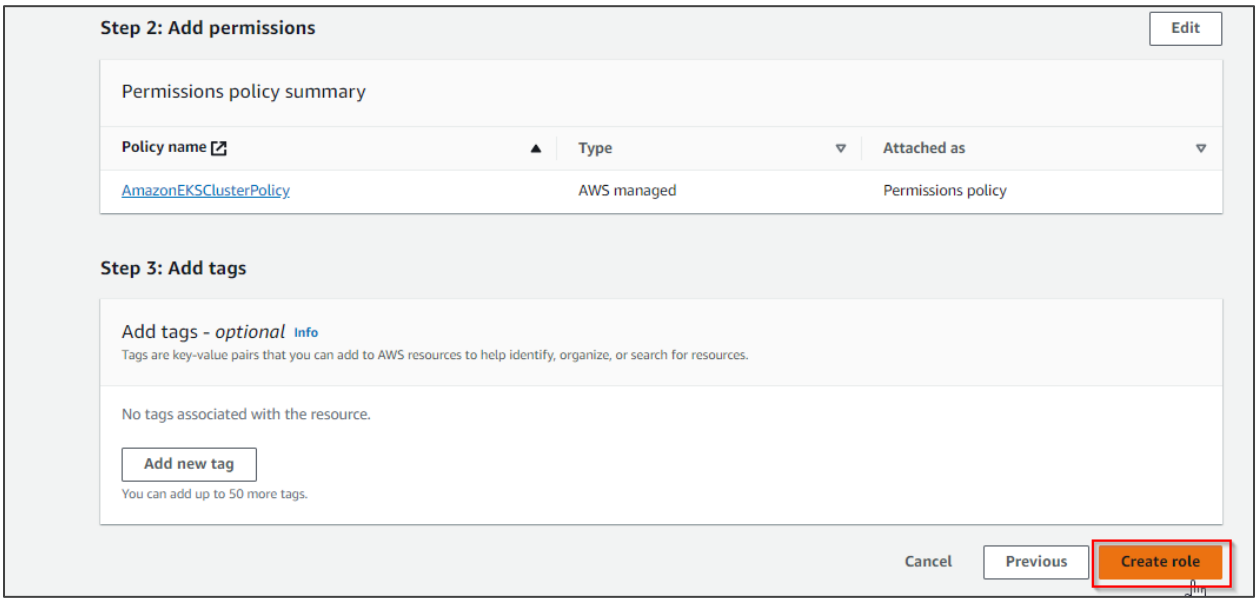
1.7 Click on the **Next** button



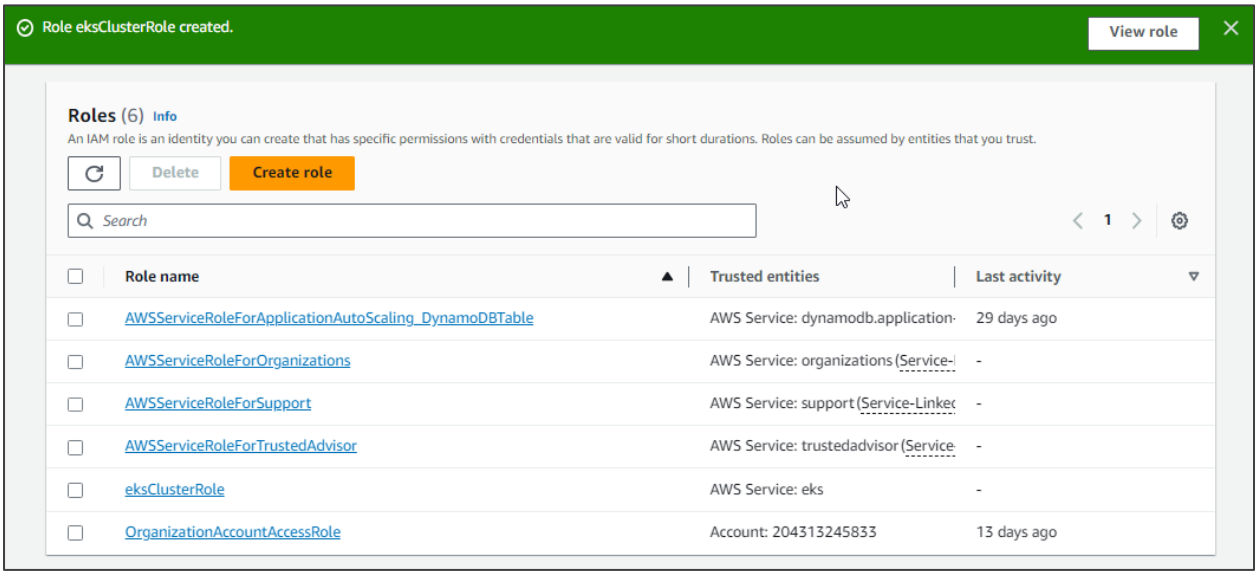
1.8 Enter the name of the role



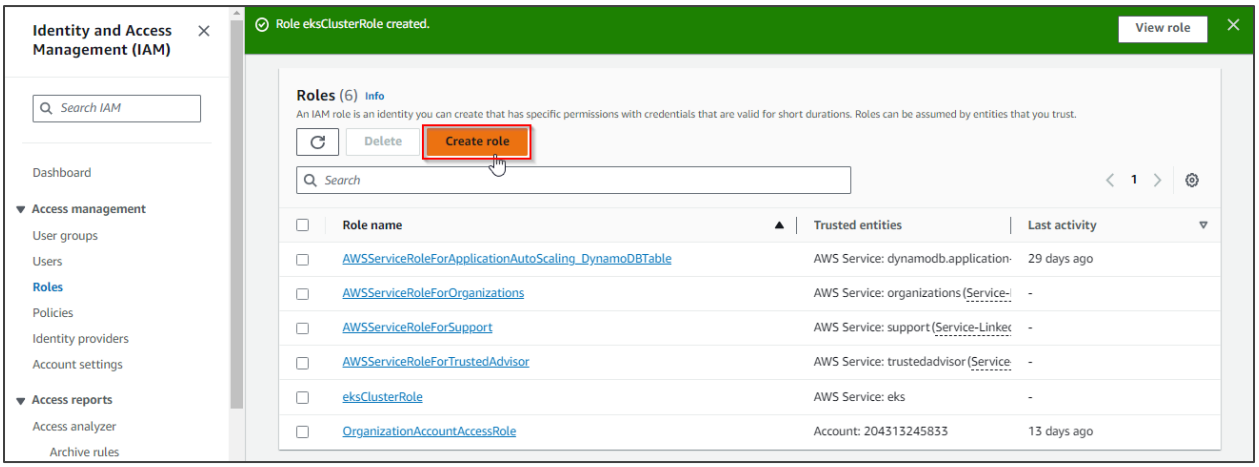
1.9 Scroll down and click on the **Create role** button



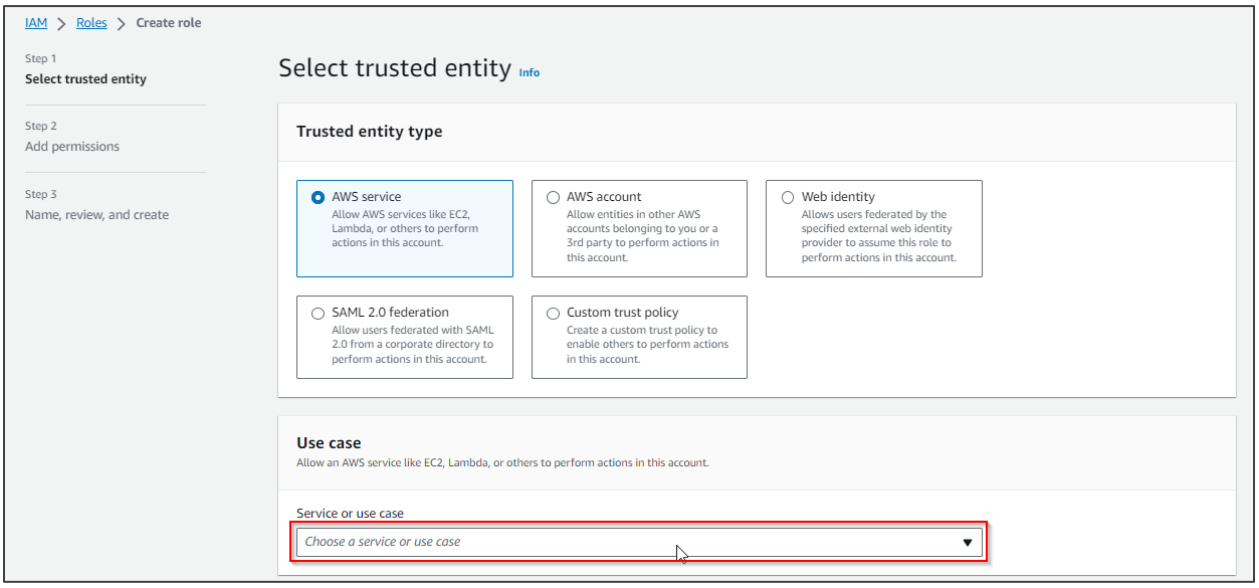
The role of the EKS cluster has been created successfully.

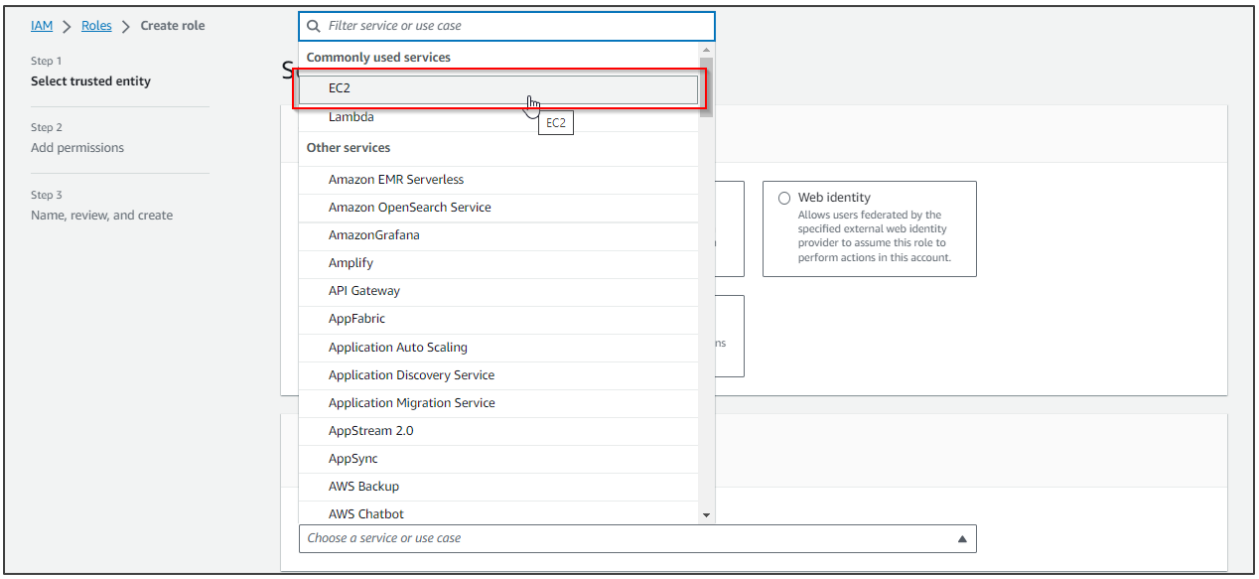


1.10 Now, click on the **Create role** button to create a role for the EKS worker node group

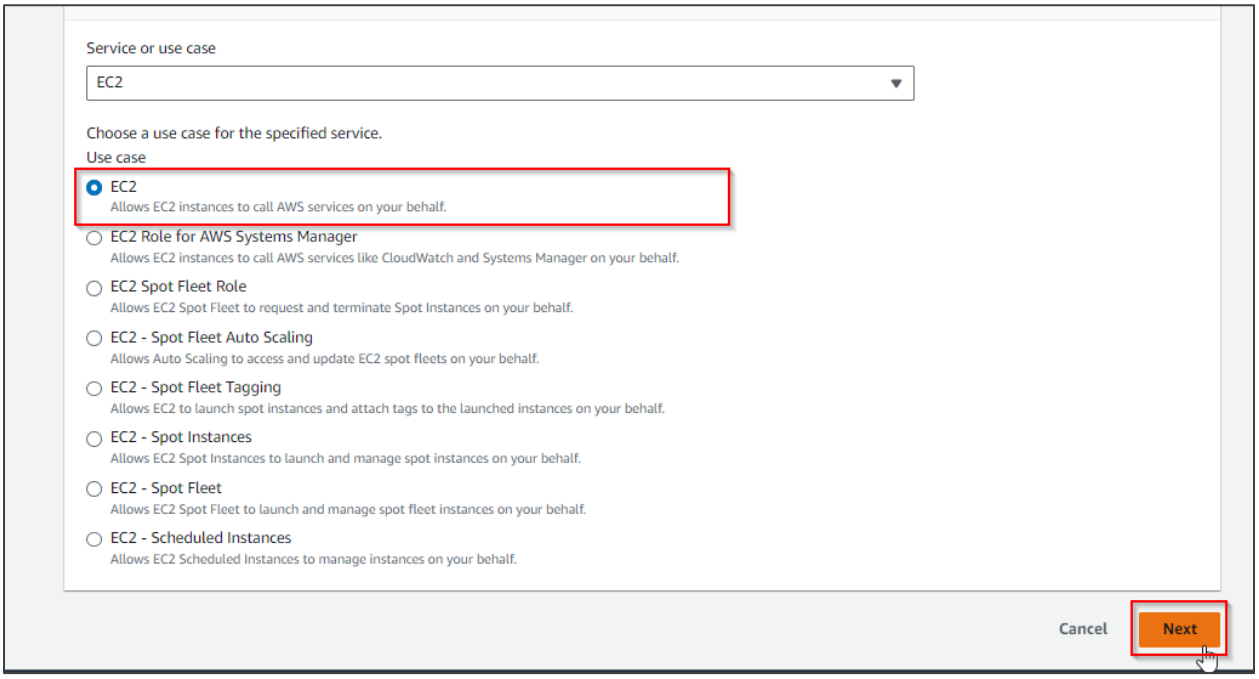


1.11 In the **Use case** section, select **EC2** service from the drop-down list

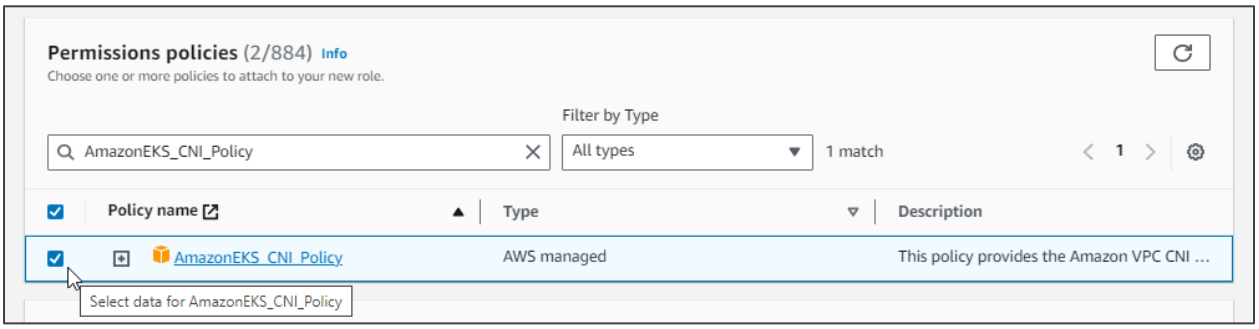


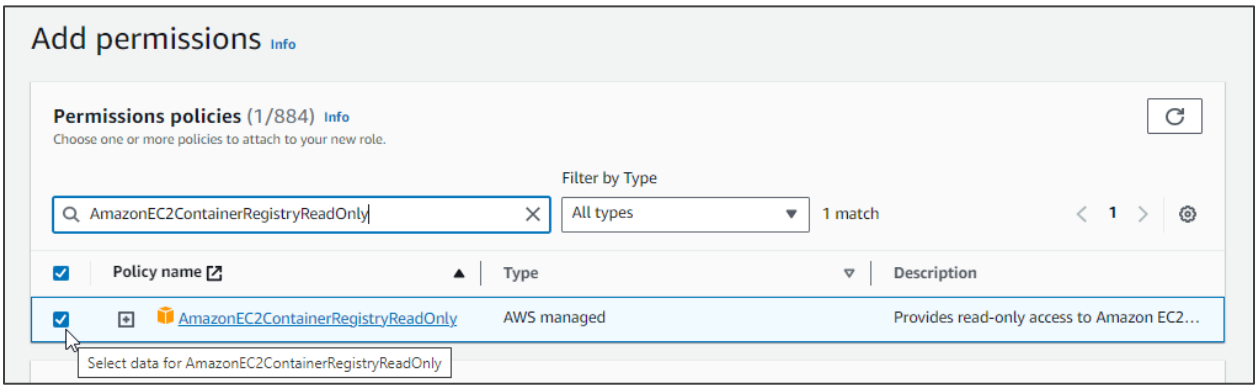


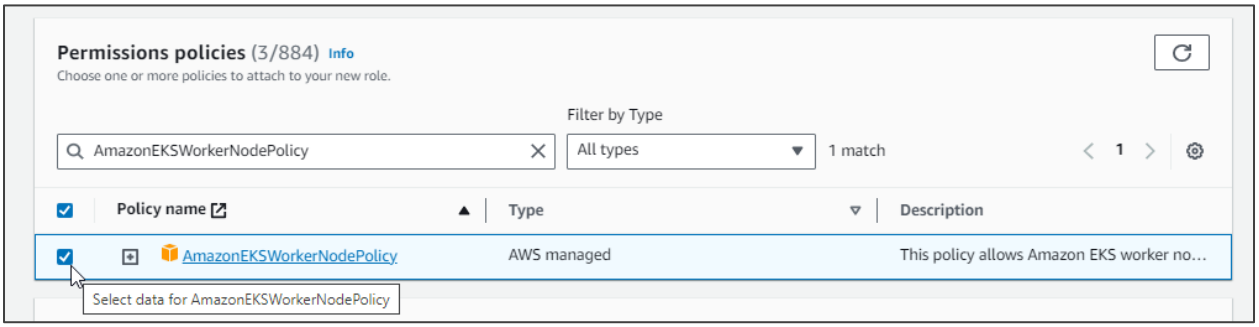
1.12 Now, select **EC2** and click on the **Next** button



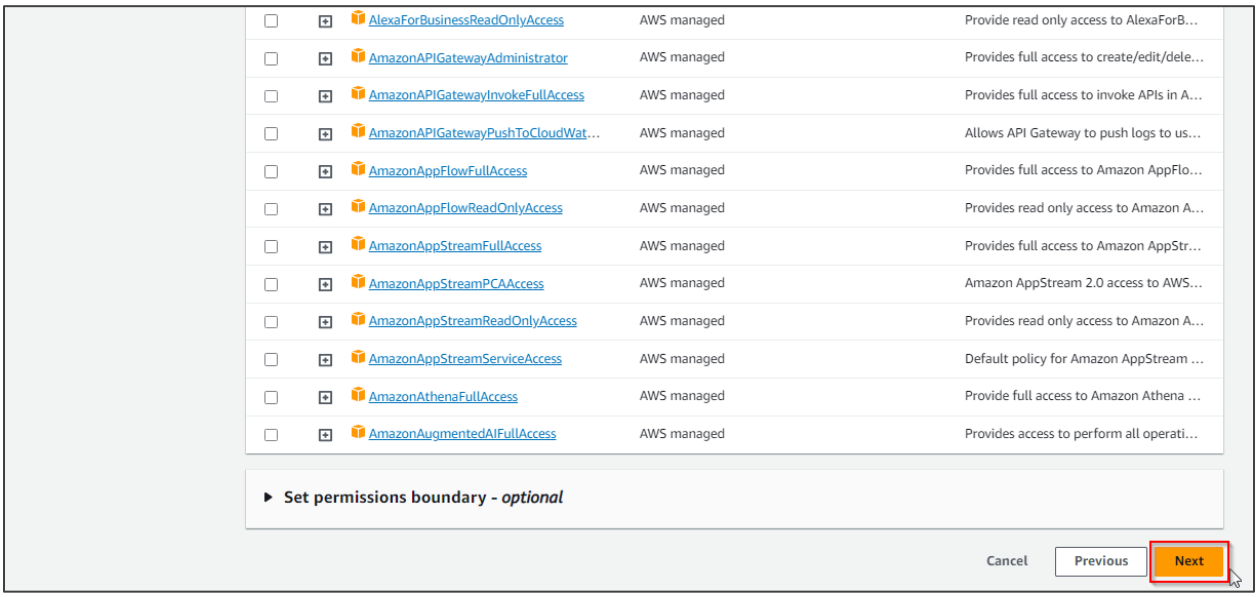
1.13 In the **Add permissions** section, search for **AmazonEKS_CNI_Policy, AmazonEC2ContainerRegistryReadOnly**, and **AmazonEKSWorkerNodePolicy,** and click on the **checkbox** to add the policies as shown in the screenshots below:







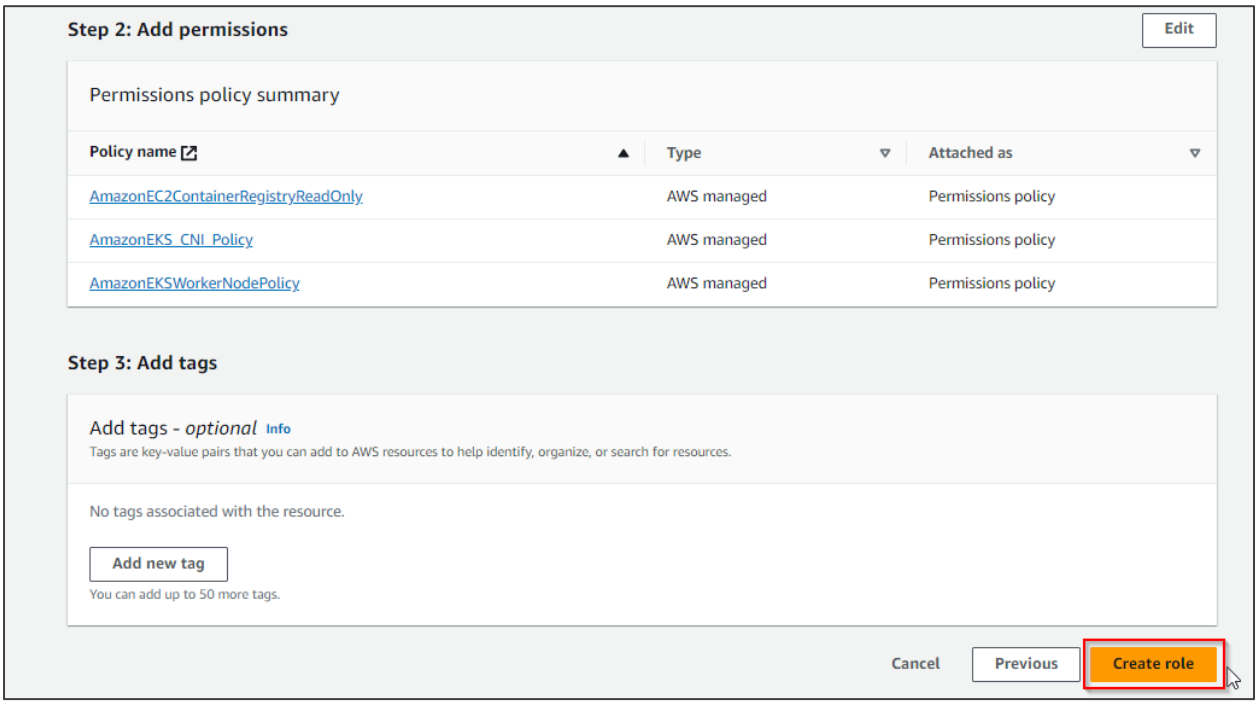
1.14 After adding the policies, click on the **Next** button



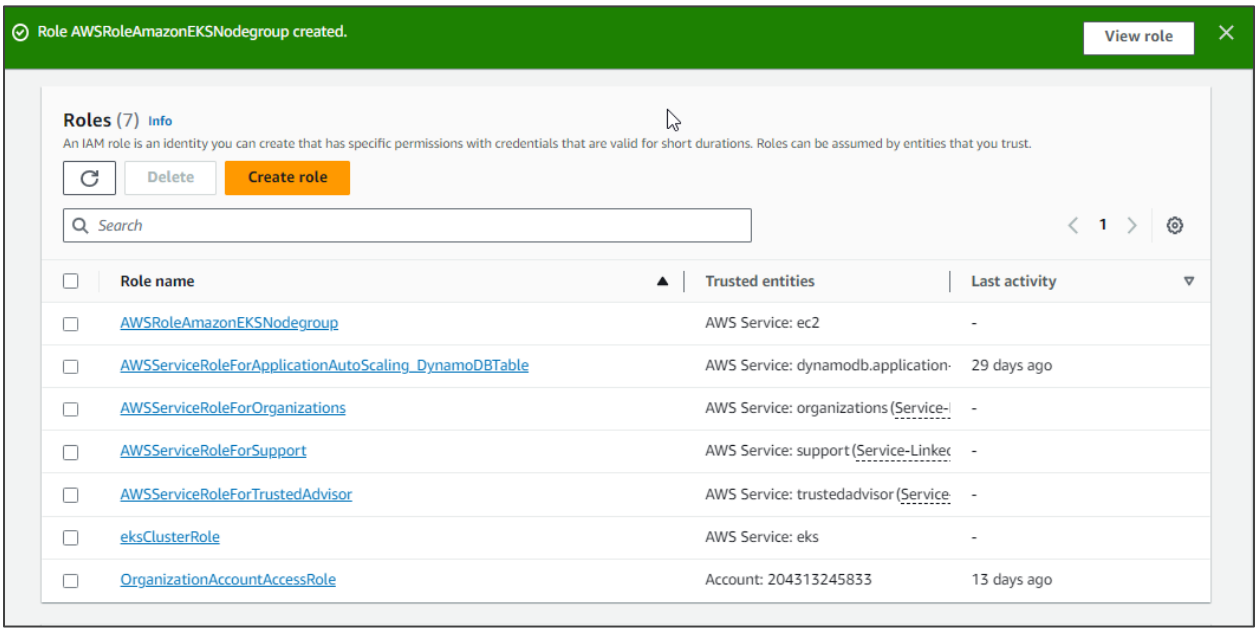
1.15 Enter the name of the role

Note: Enter a unique name for the role, such as **AWSRoleAmazonEKSNodegroup**

1.16 Click on the **Create role** button

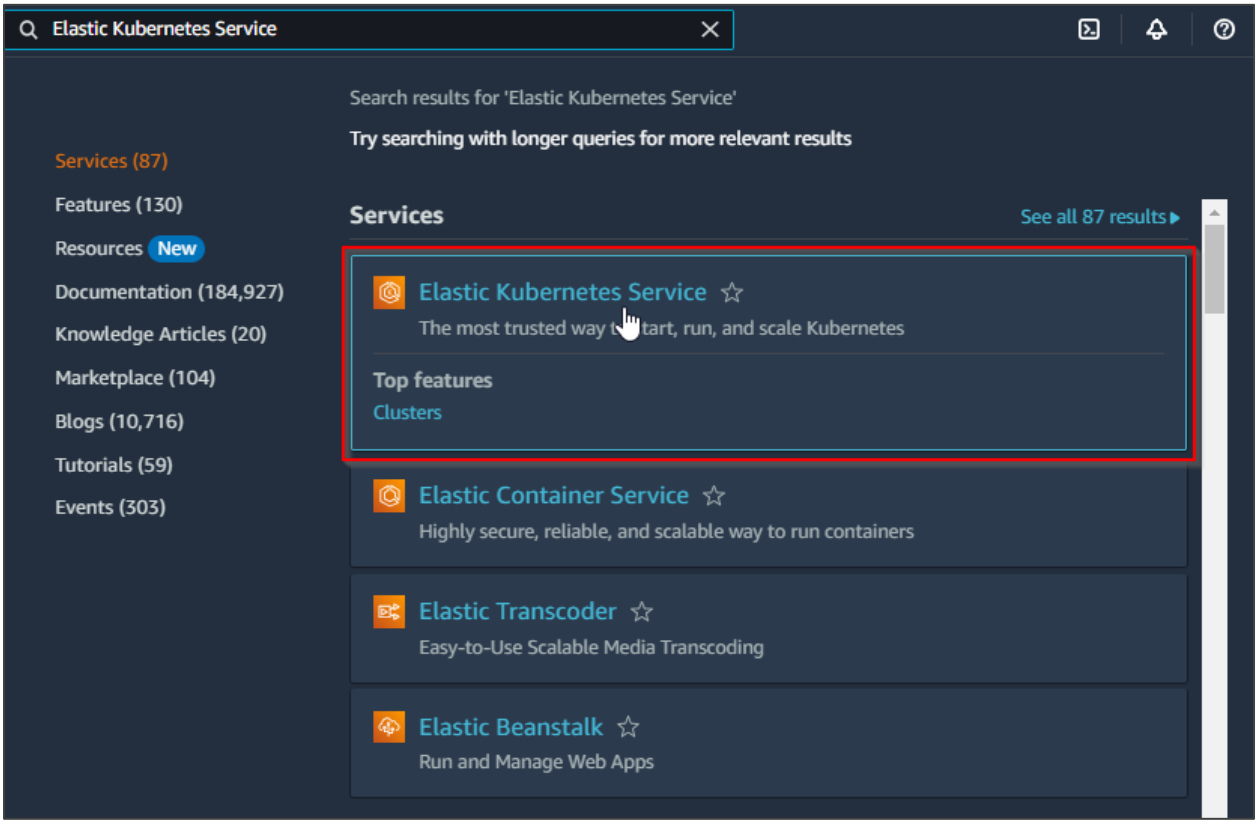


The role of the EKS worker node group has been created successfully.

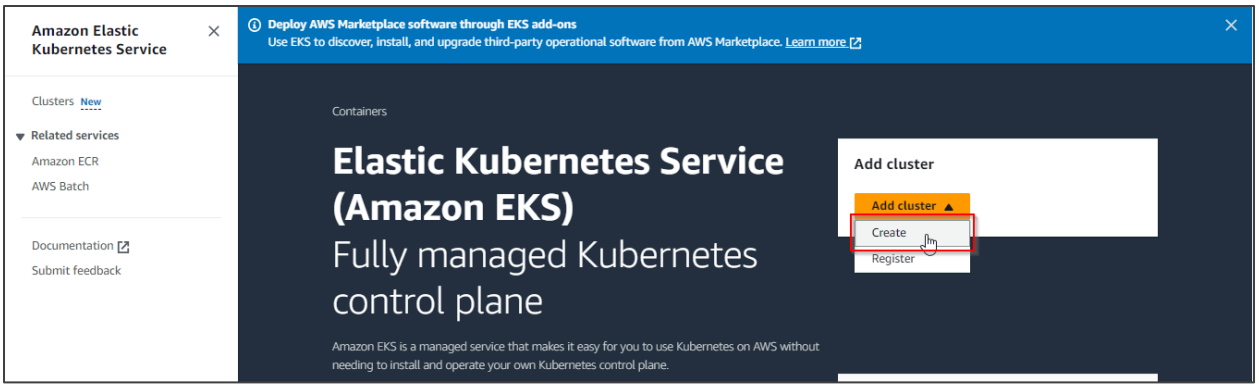


Step 2: Create the AWS EKS cluster

2.1 Search for and click on **Elastic Kubernetes Service**

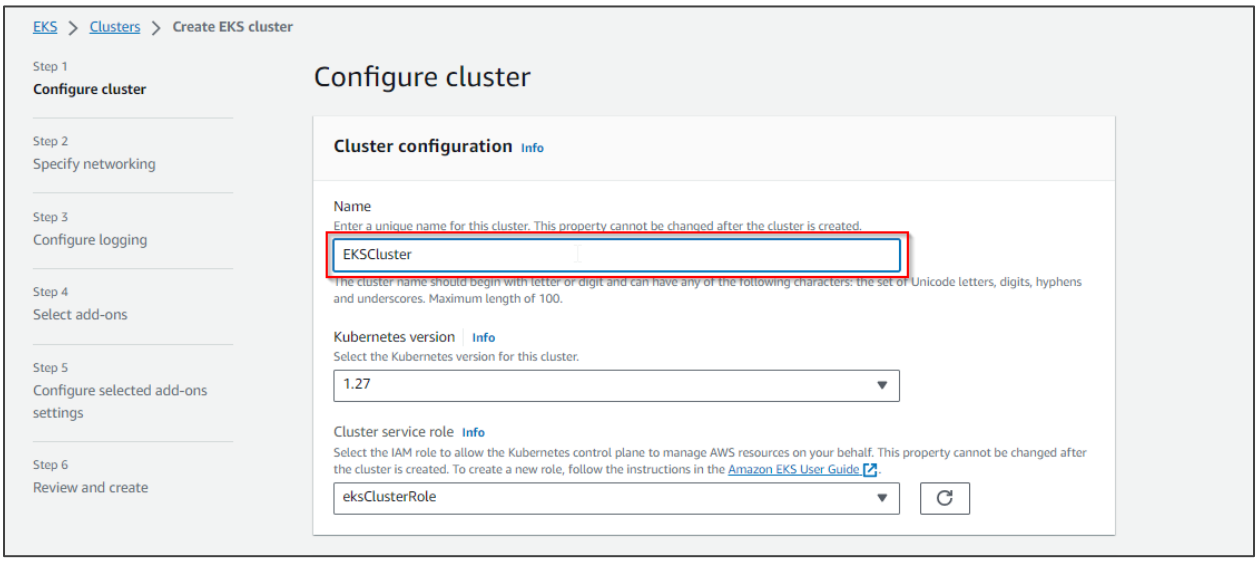


2.2 Click on the **Add cluster** button and then click on **Create**

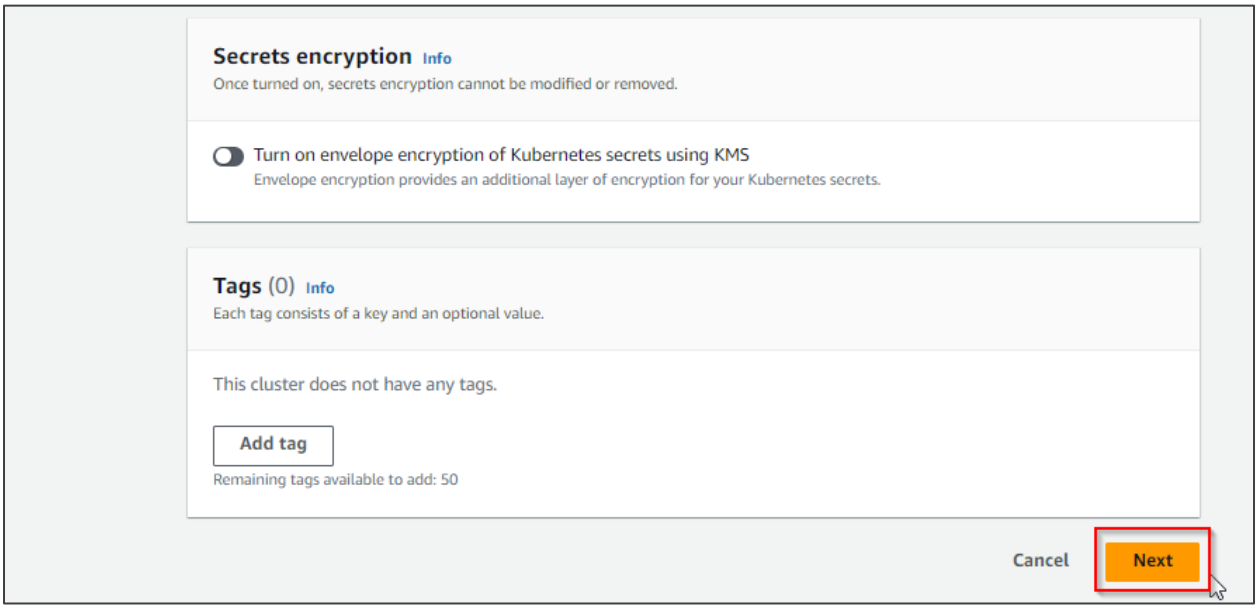


2.3 Enter the name of the cluster

Note: You may select the Kubernetes version as required, and the cluster service role will automatically load since we have already created the IAM role for the same.

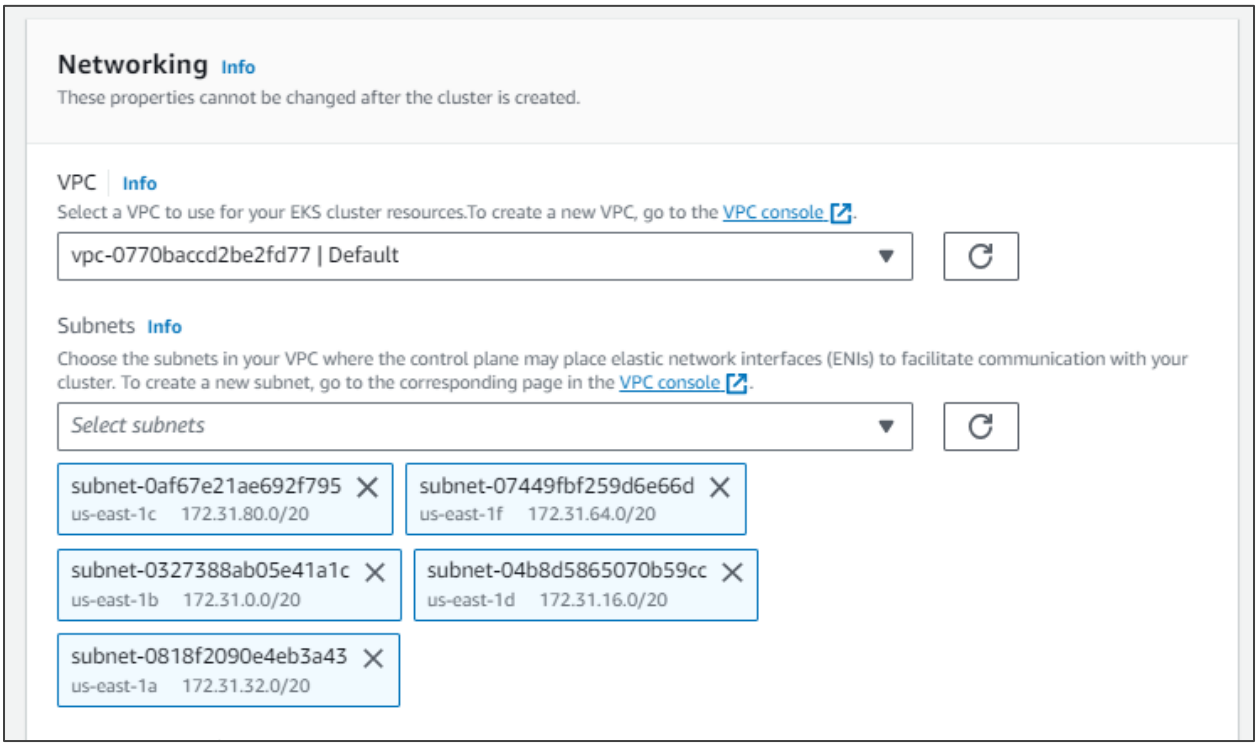


2.4 Click on the **Next** button

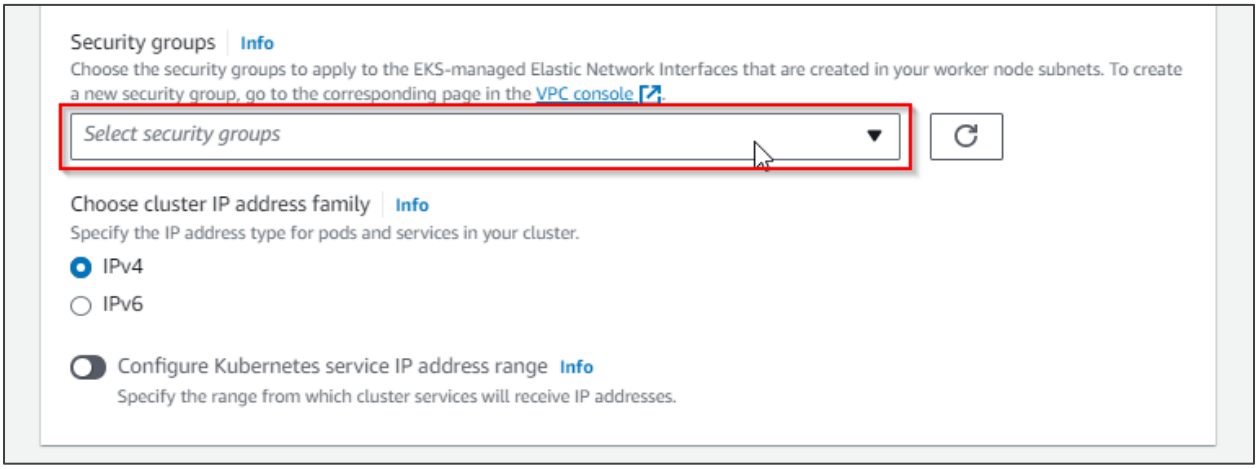


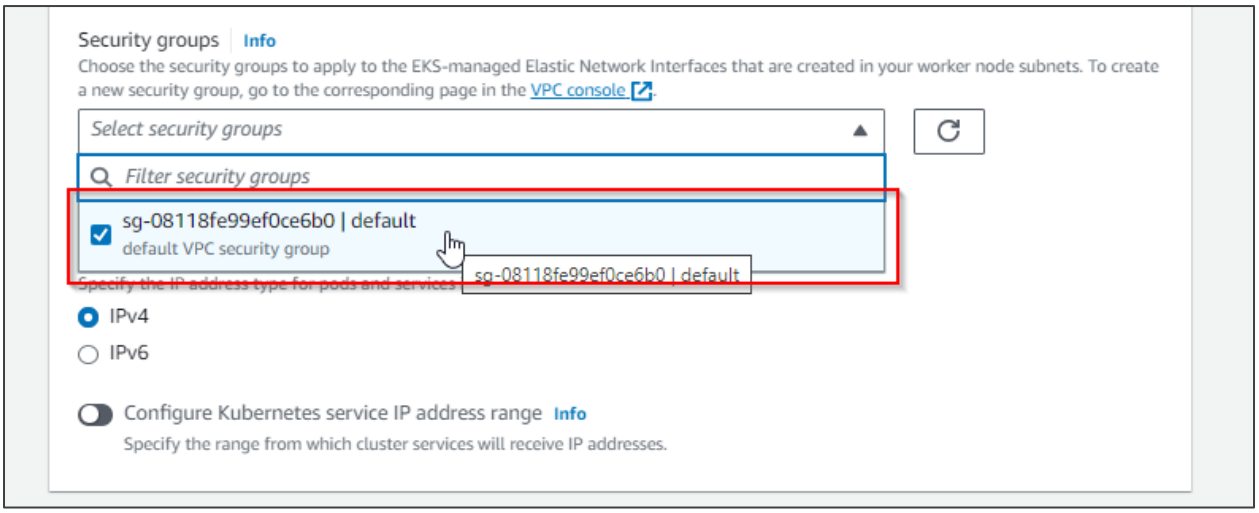
2.5 Provide networking configuration such as **VPC** and **Subnets**

Note: You need to unselect **us-east-1e**; the targeted availability zone does not currently have sufficient capacity to support the cluster.

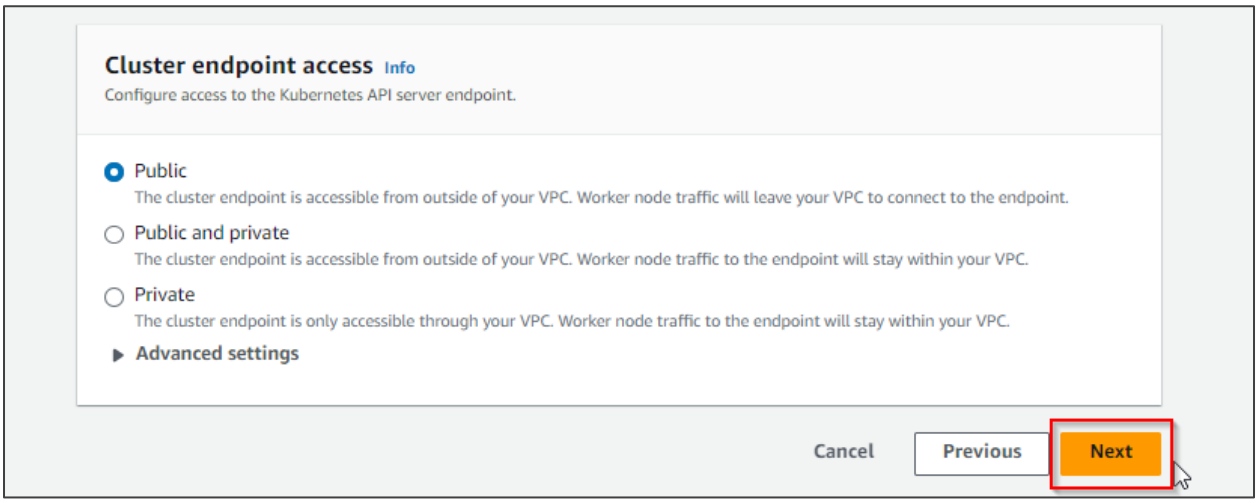


2.6 In the **Security groups** section, select the **default VPC security group**

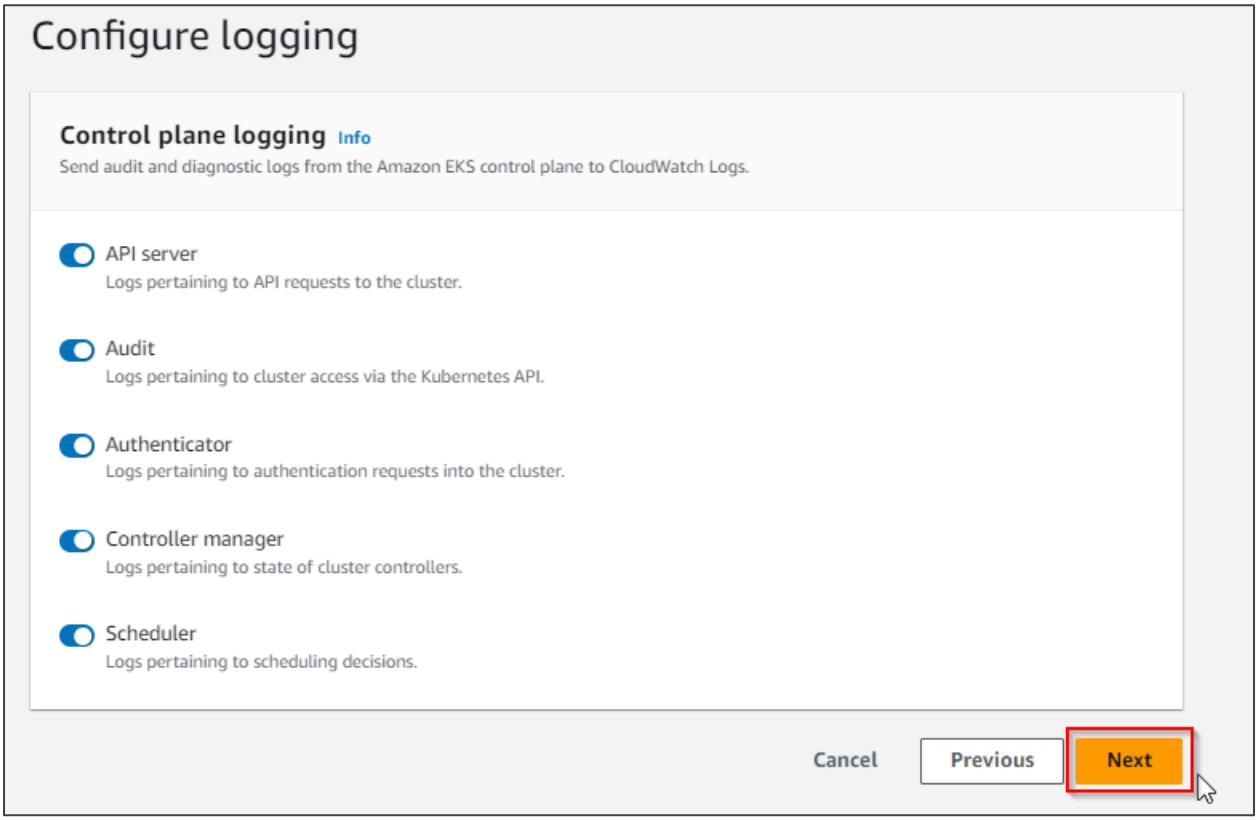




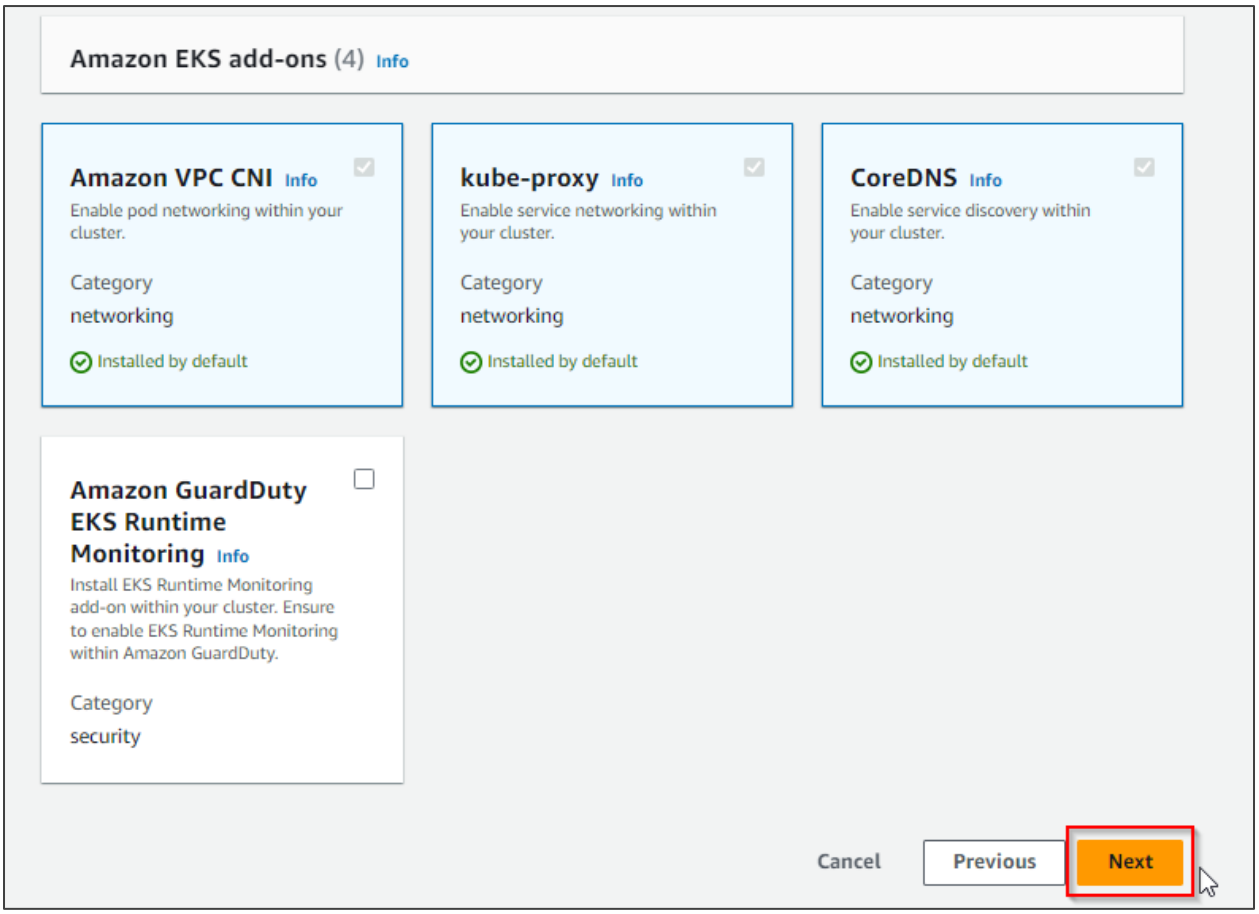
2.7 Configure cluster endpoint access to specify the access and click on the **Next** button



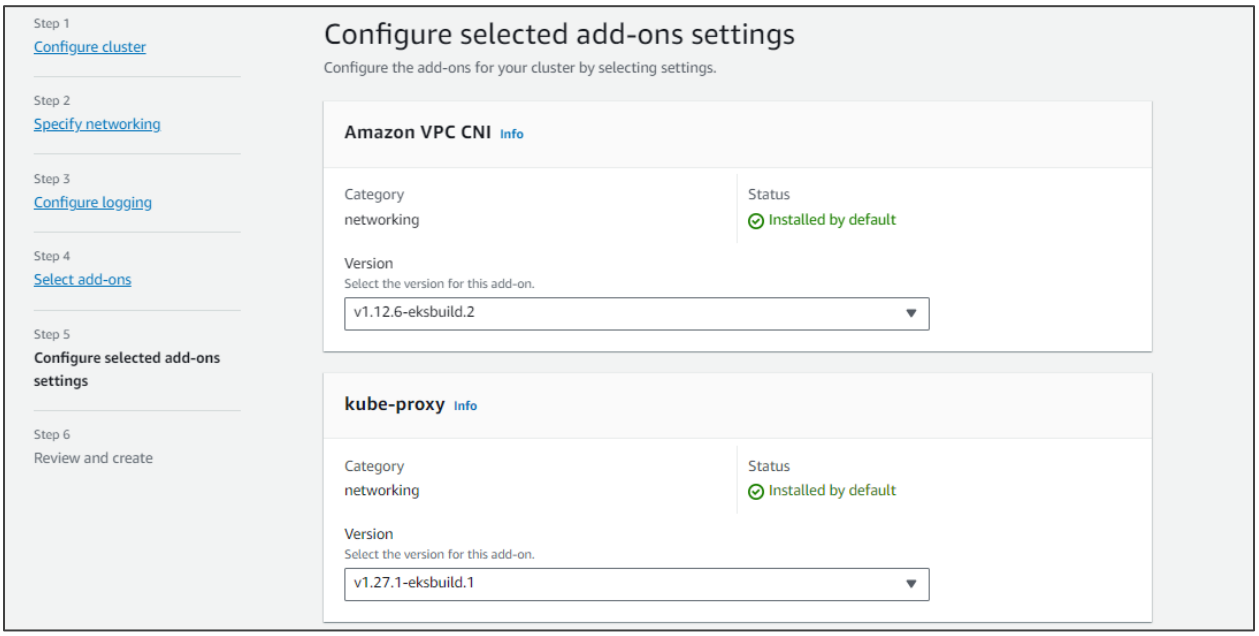
2.8 Configure the EKS Cluster logging for the control plane component as shown in the screenshot below, and click on the **Next** button

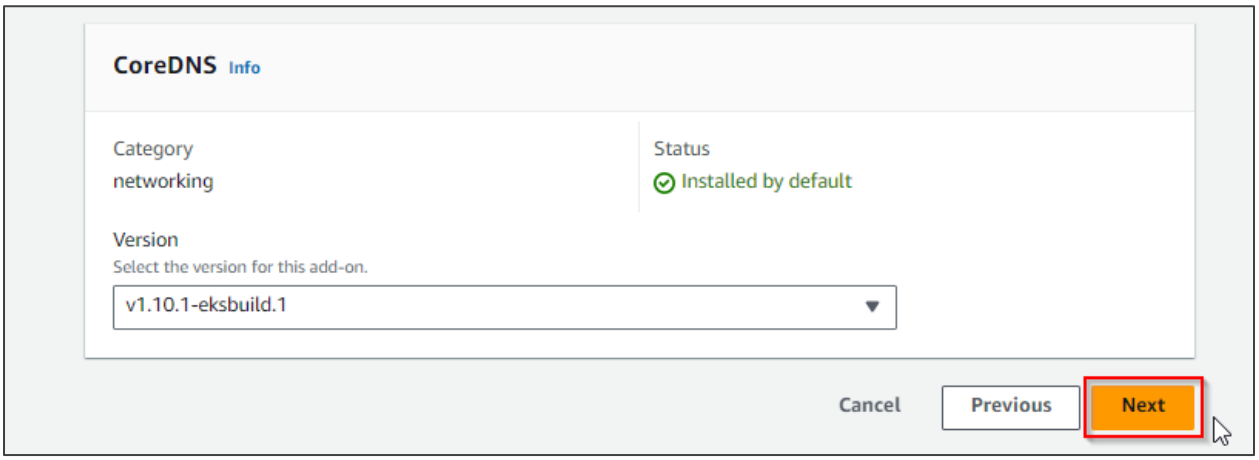


2.9 Select the required add-ons and click on the **Next** button

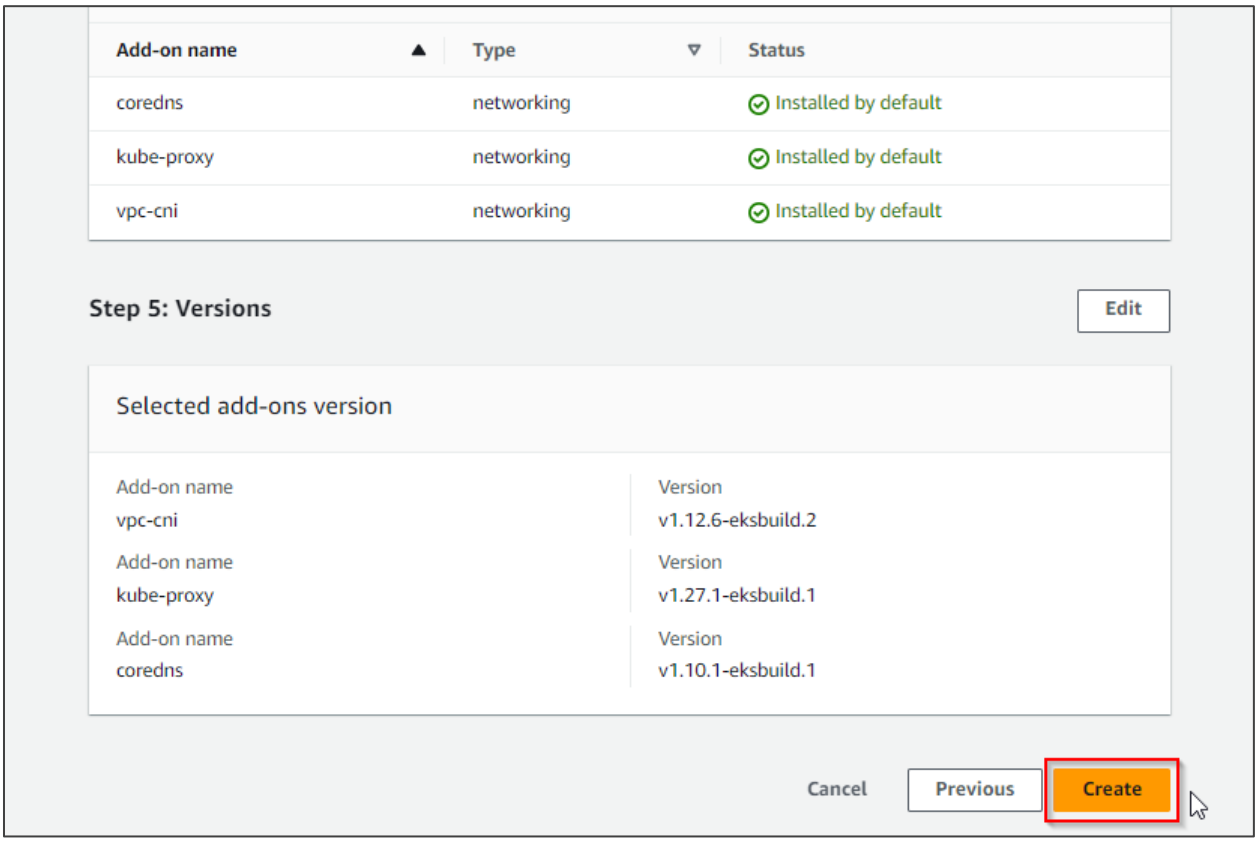


2.10 Now, configure the add-ons per the requirement, and click on the **Next** button

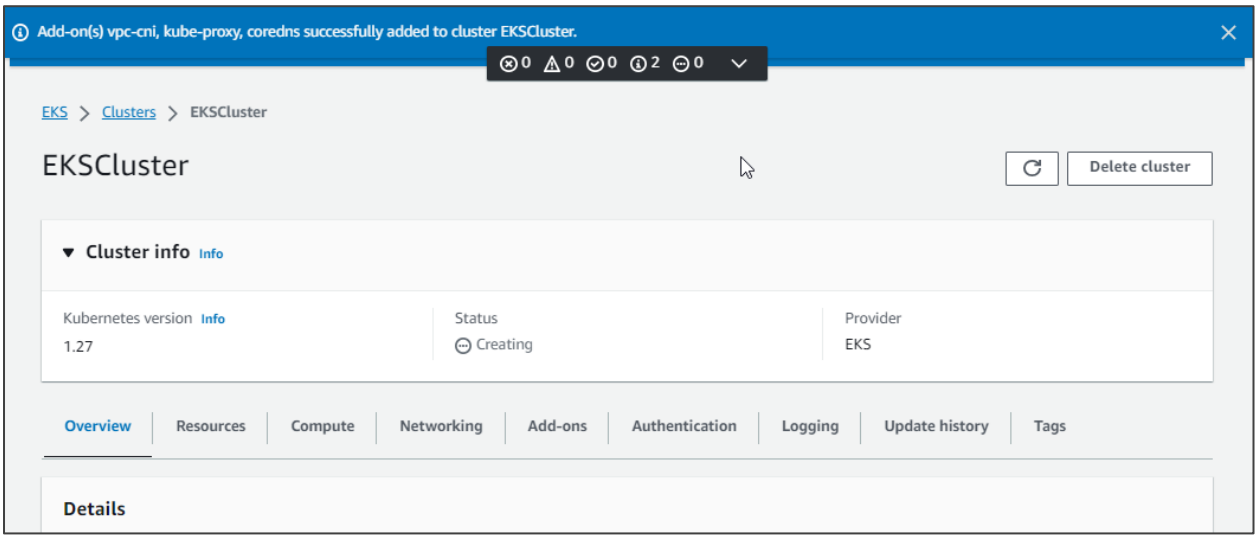




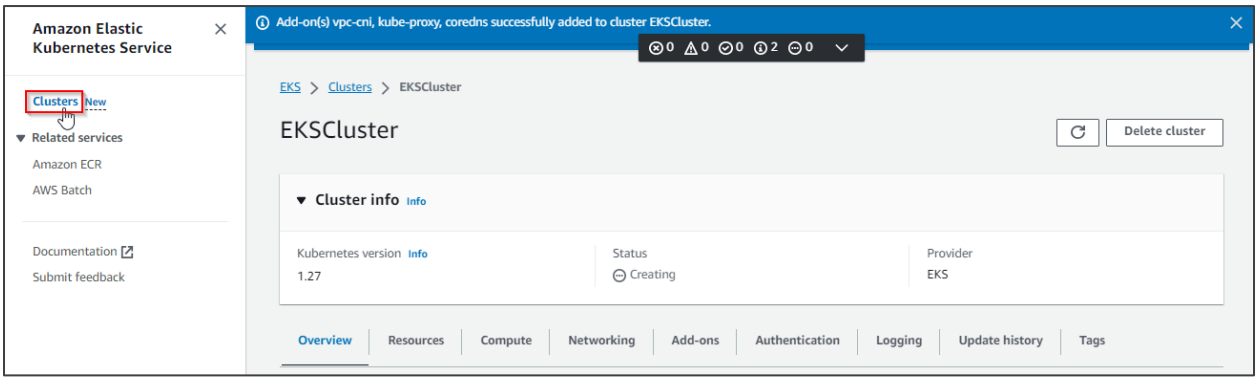
2.11 Scroll to the bottom of the page and click on the **Create** button



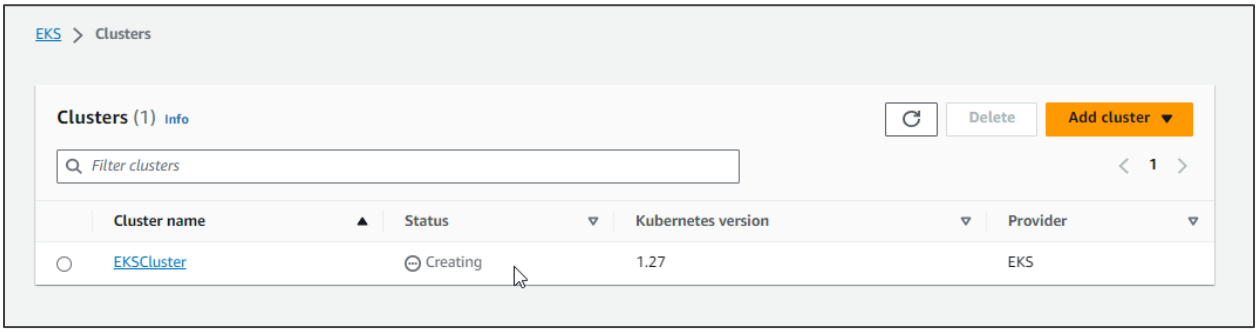
The cluster has been created successfully.

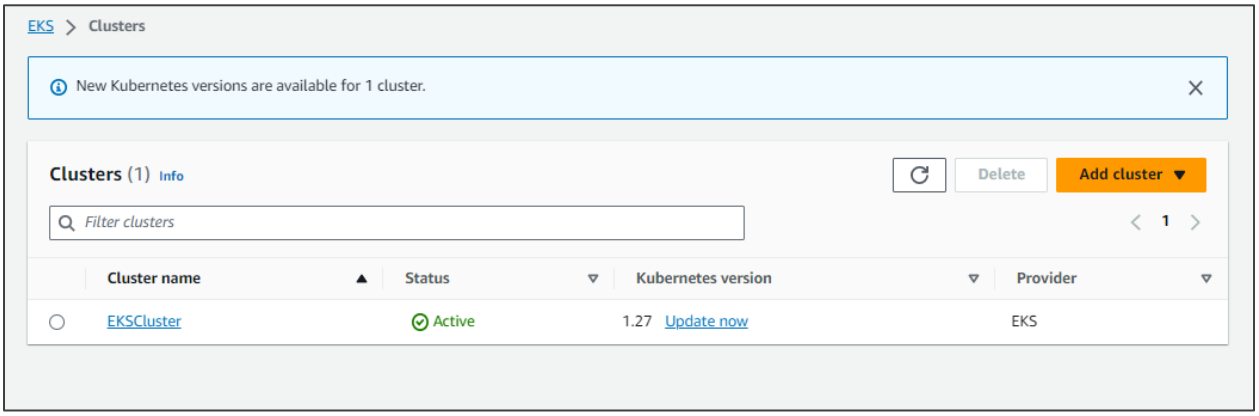


2.12 Click on the **Clusters** to validate its availability, as shown in the screenshot below:



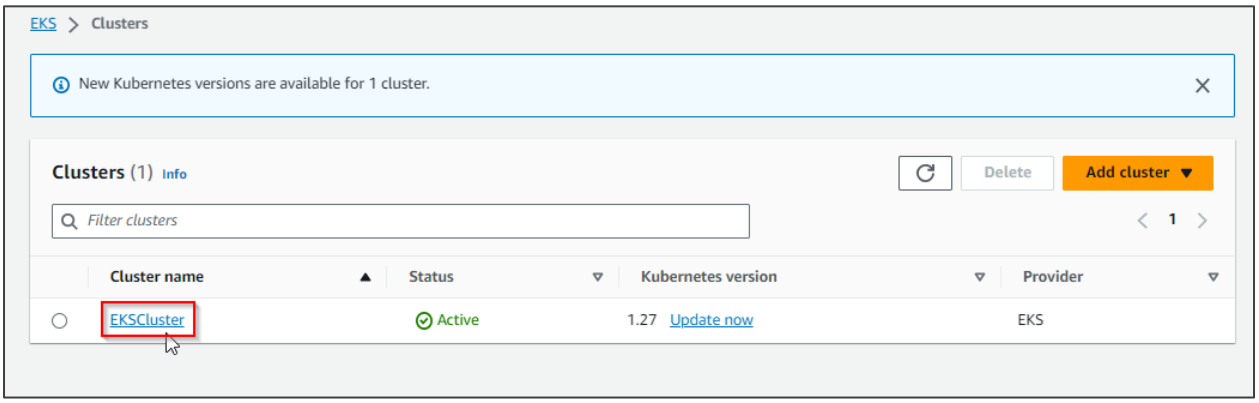
It takes a few minutes for the cluster to be active.



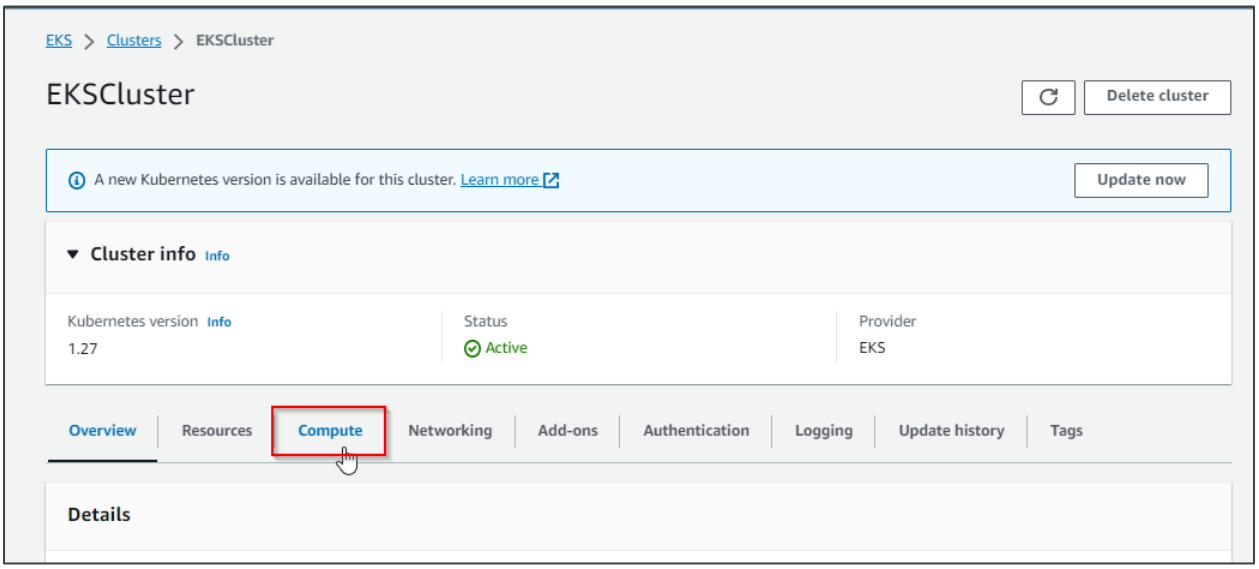


Step 3: Create the AWS EKS worker node group

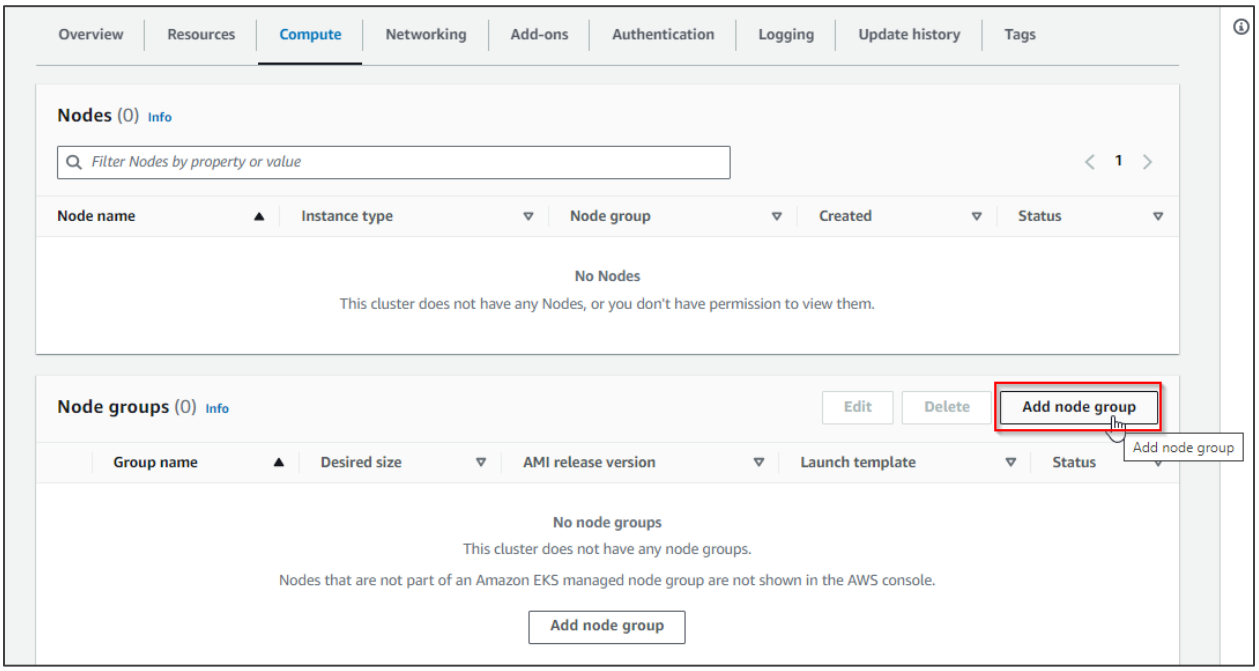
3.1 Click on the created cluster as shown in the screenshot below:



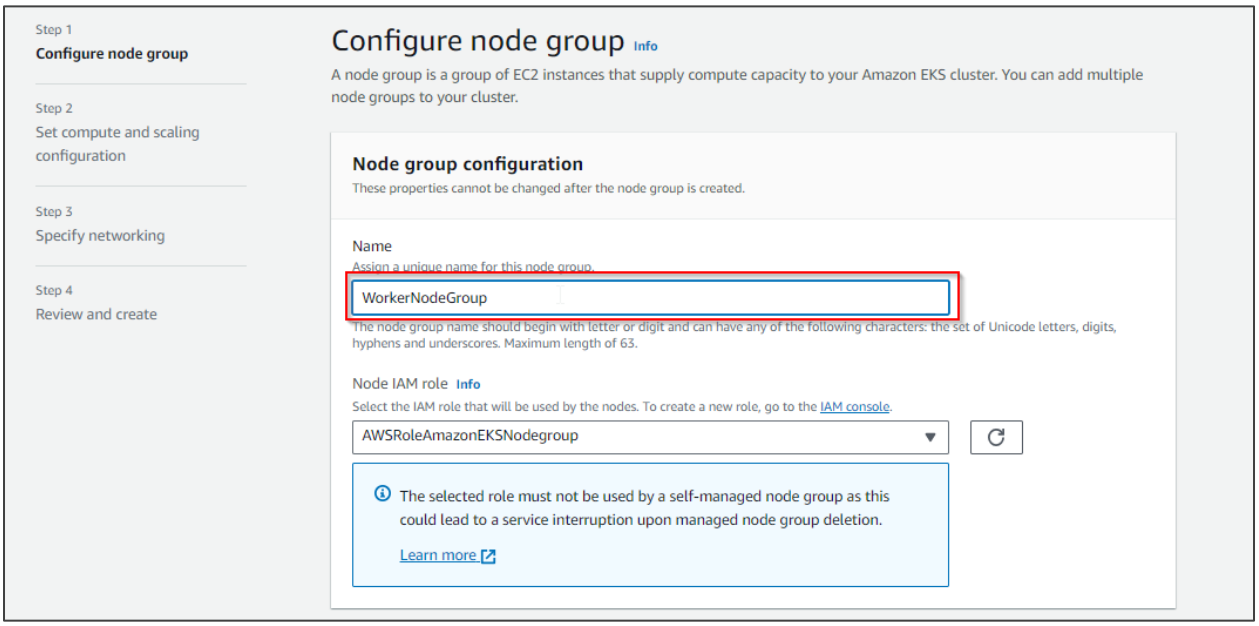
3.2 Click on **Compute**



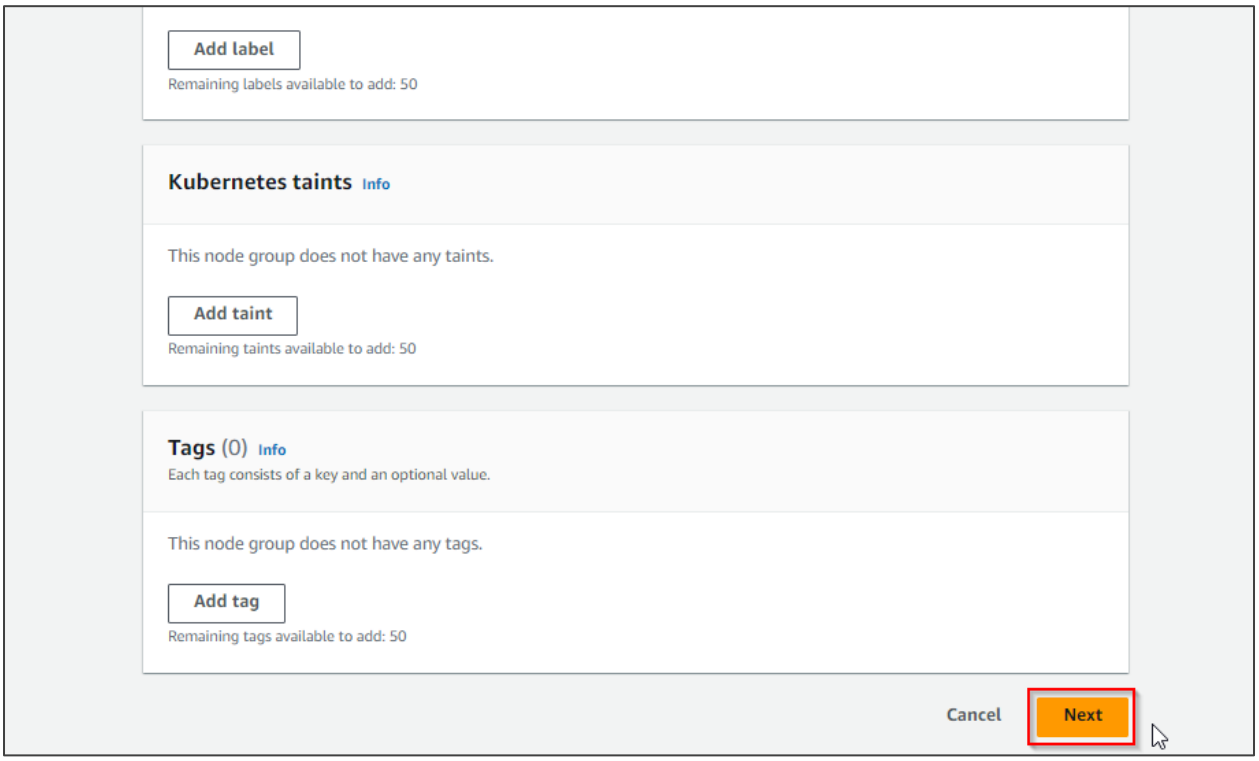
3.3 Click on the **Add node group** button



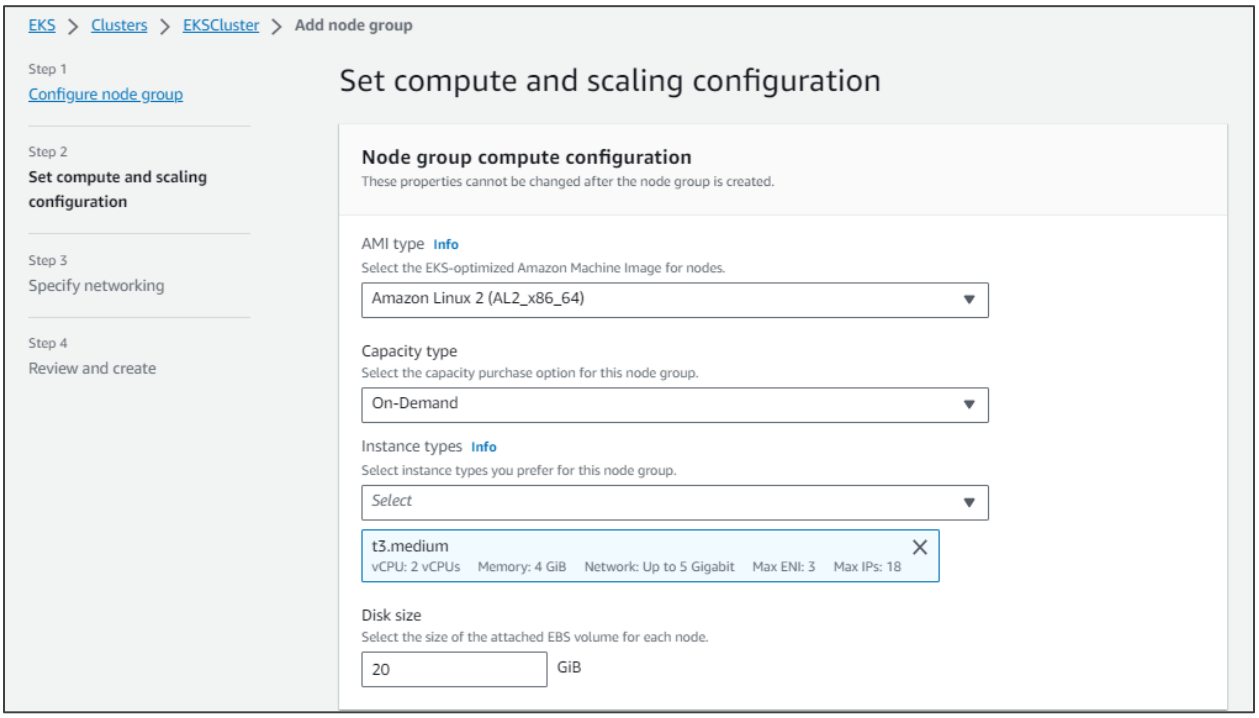
3.4 Add the name for the node group



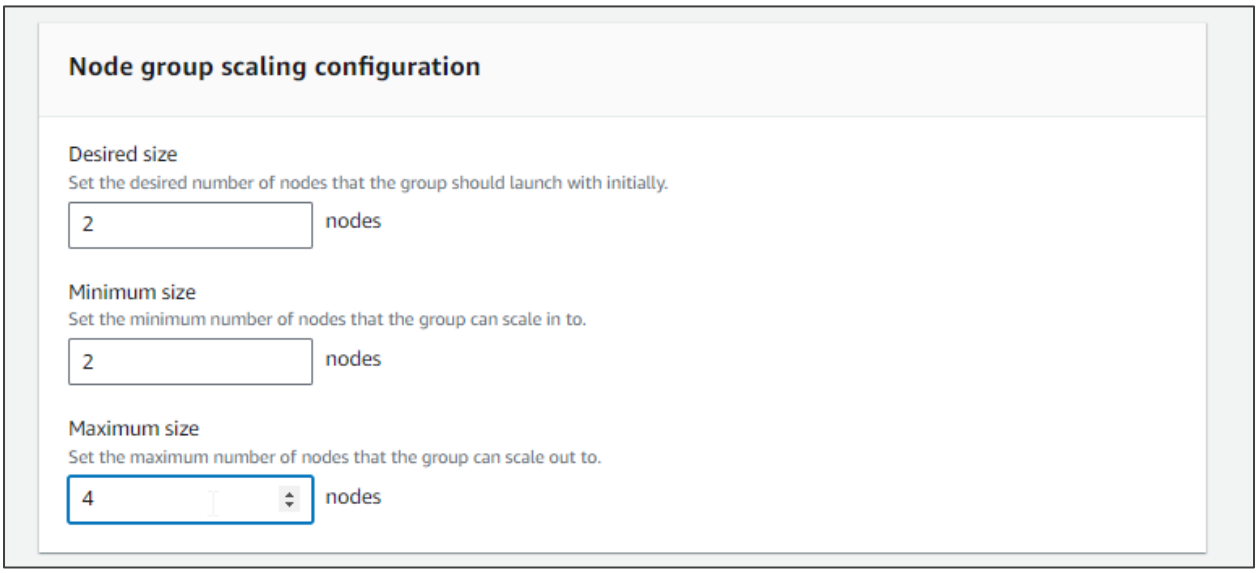
3.5 Scroll to the bottom of the page and click on the **Next** button



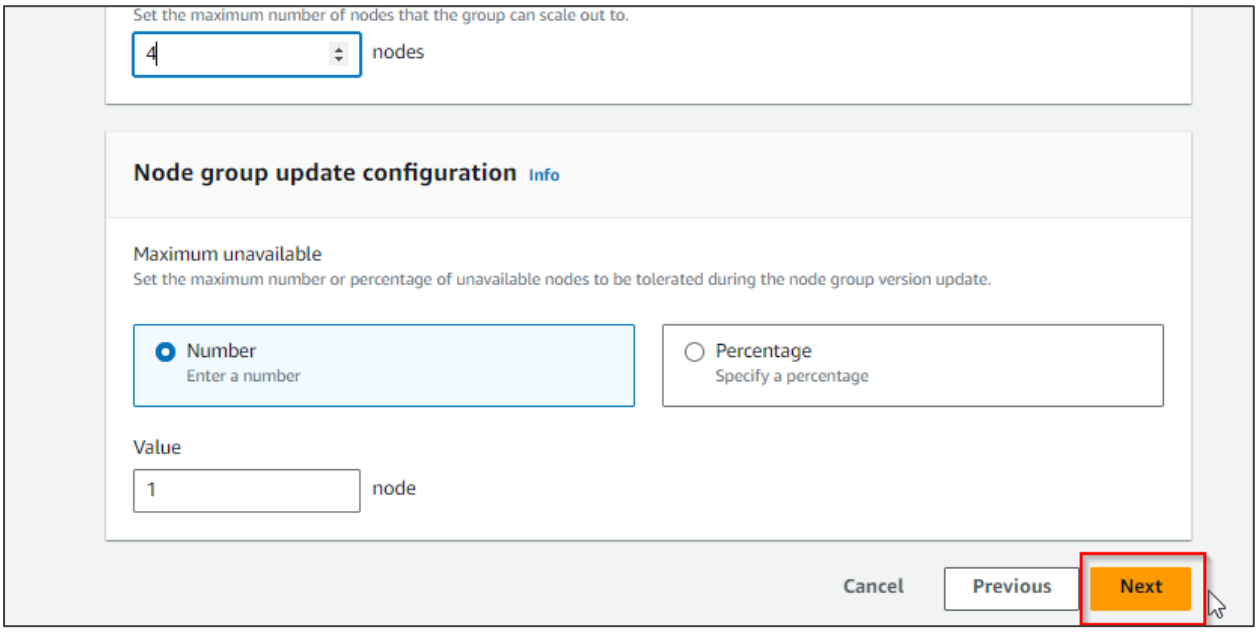
3.6 Provide compute and scaling configuration for worker nodes



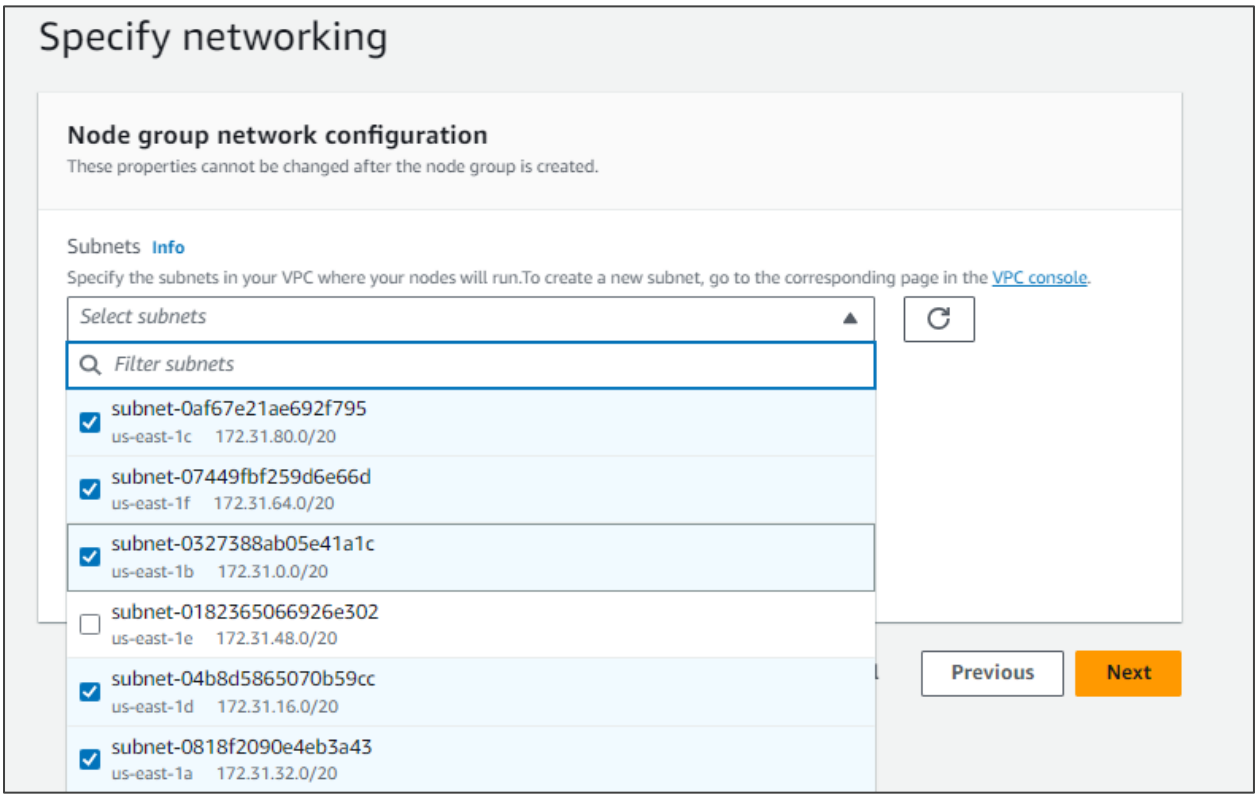
3.7 Now, provide Node scaling configuration with values for desired, minimum, and maximum node sizes



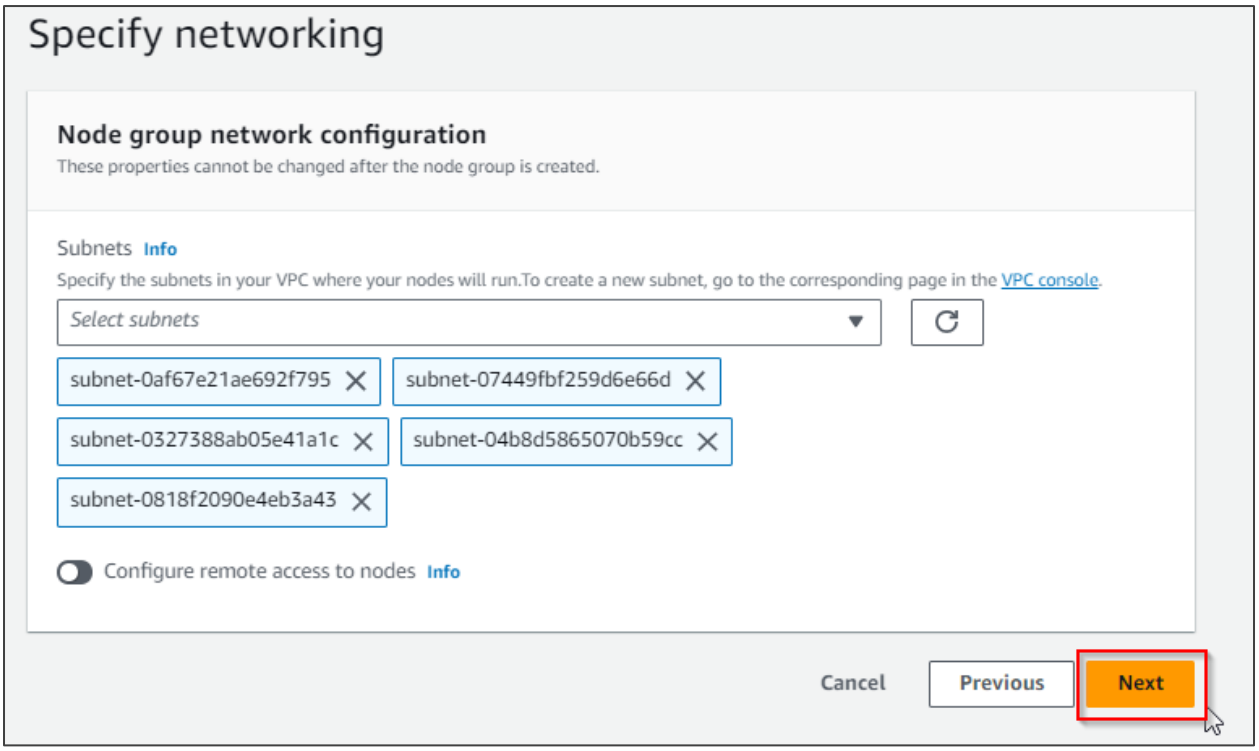
3.8 Click on the **Next** button



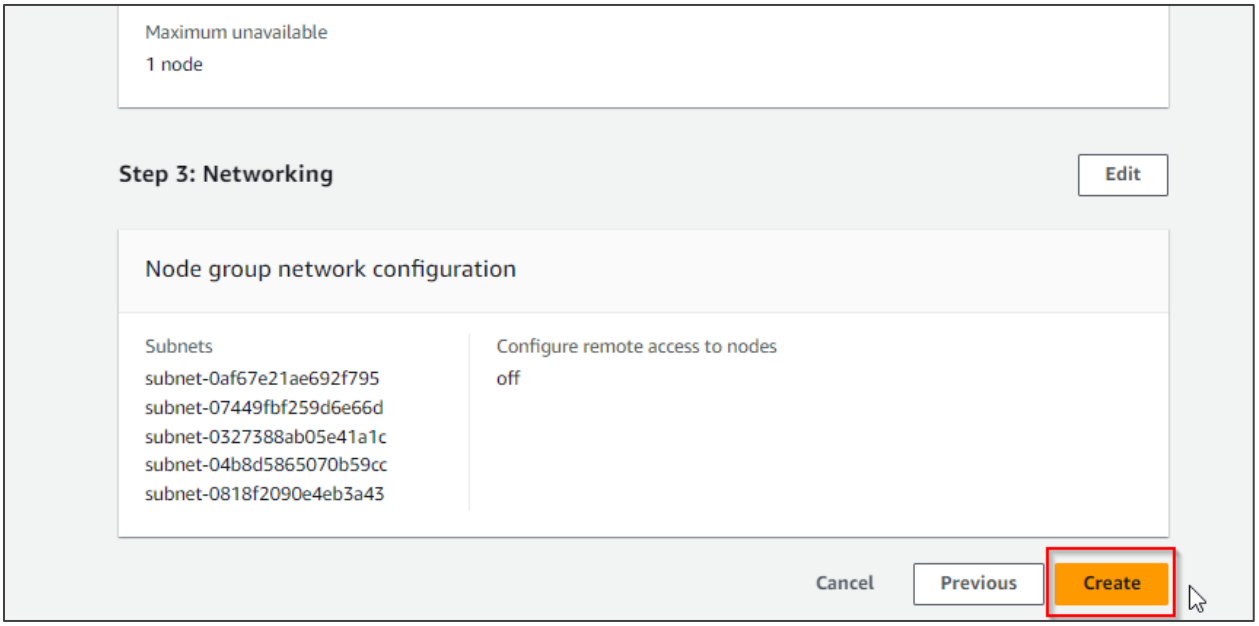
3.9 Next, choose subnets except for **us-east-1e**, as shown in the screenshot below:



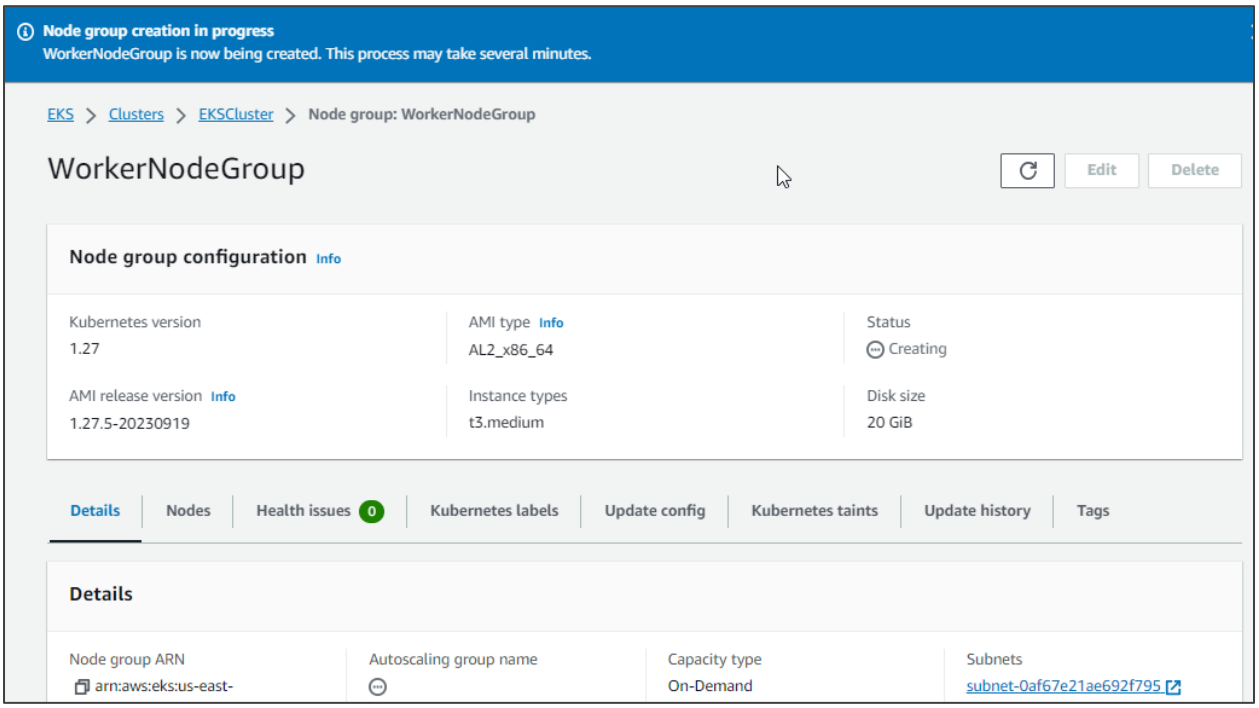
3.10 Click on the **Next** button



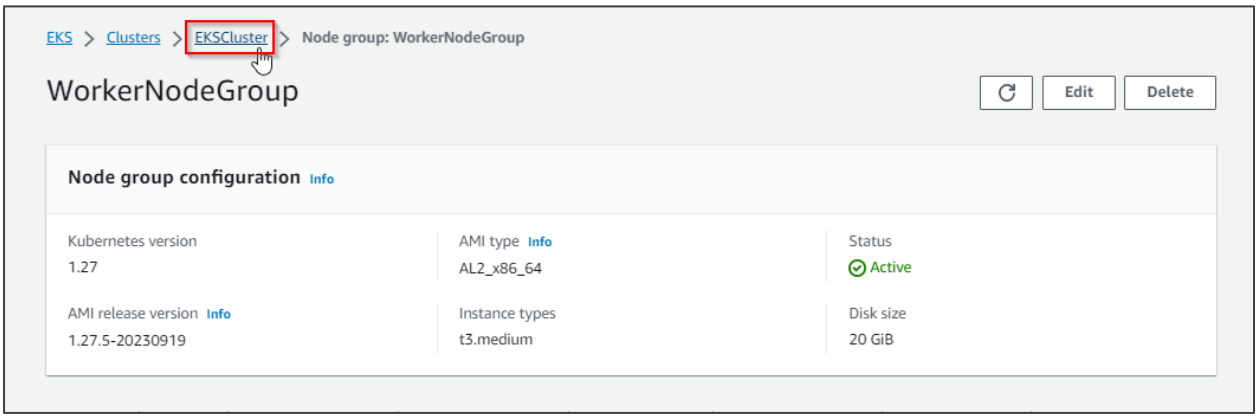
3.11 Scroll to the bottom of the page and click on the **Create** button



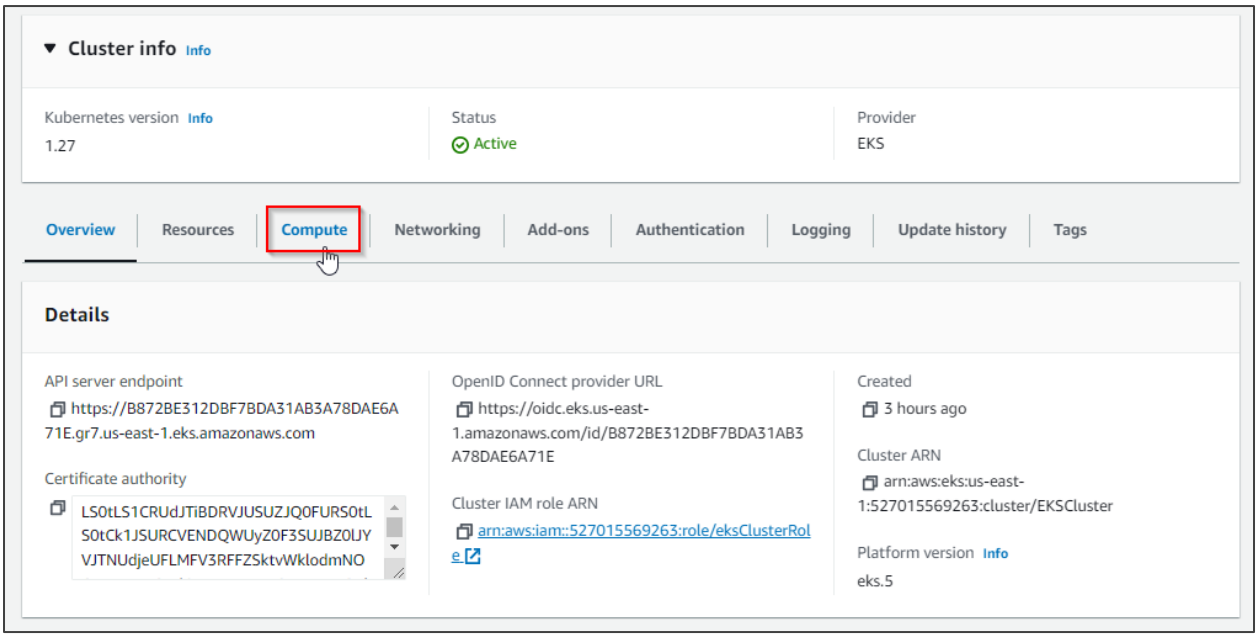
You will see the interface as shown below:



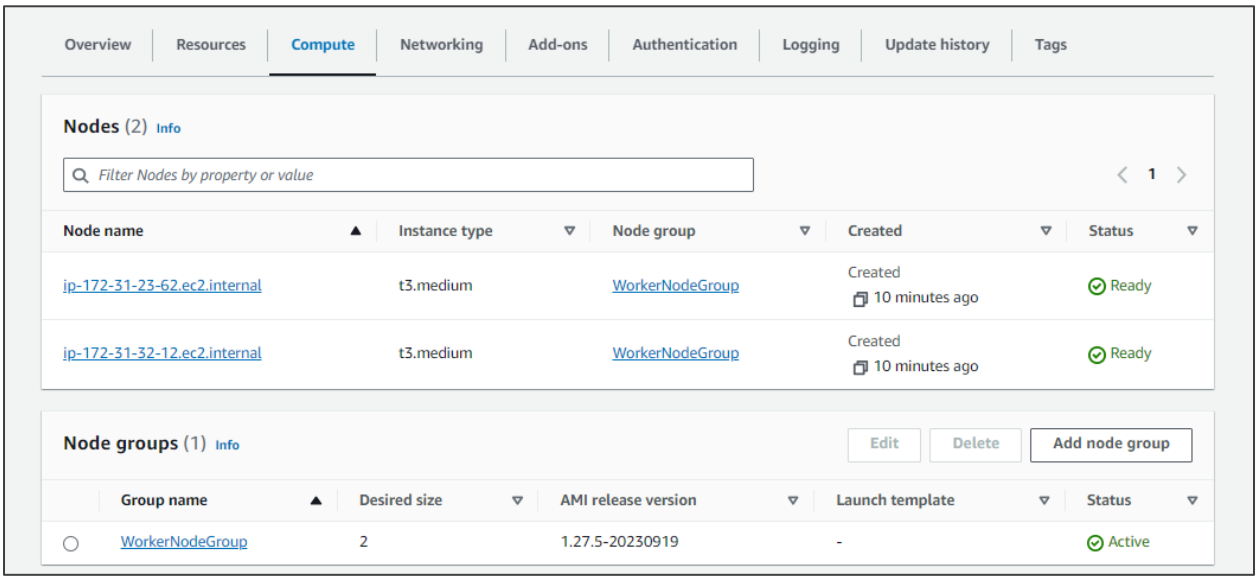
3.12 Click on **EKSCluster**



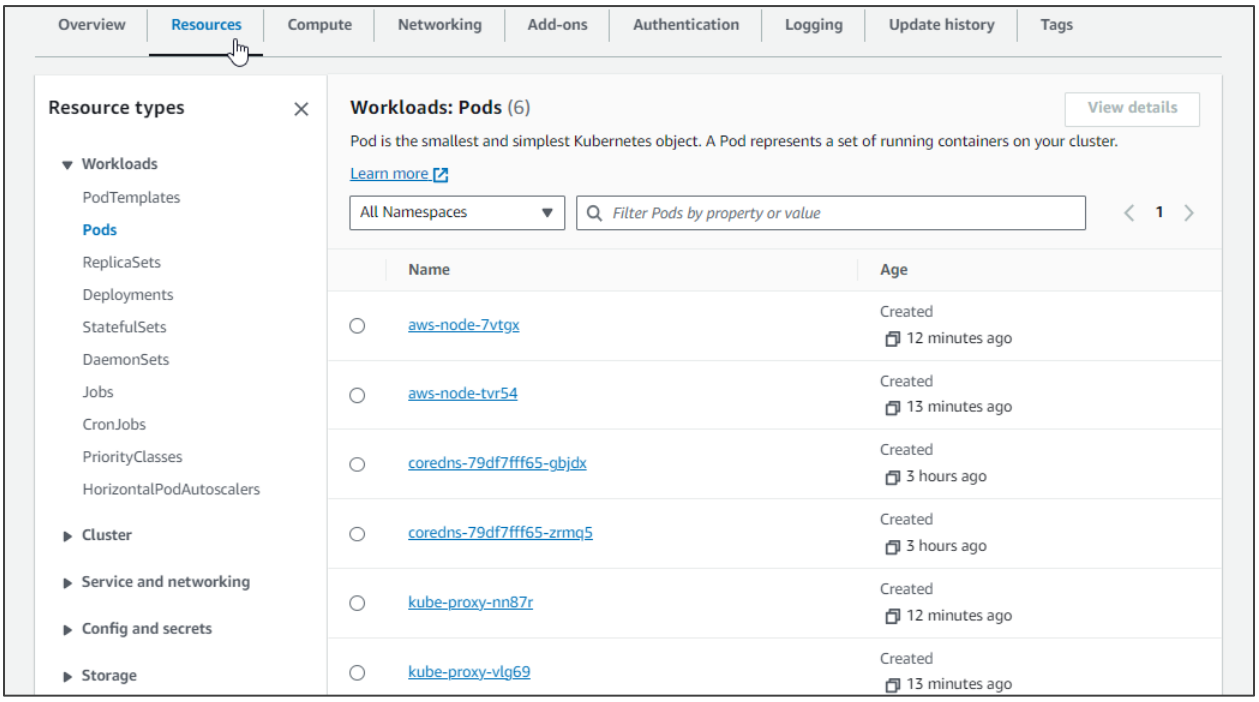
3.13 Now, click on the **Compute** tab



Once the worker node group becomes active, check the **Nodes** section to confirm that the worker nodes are ready, as shown in the screenshot below:



3.14 Click on the **Resources** tab



By following these steps, you have successfully set up and optimized an AWS EKS cluster with IAM roles, networking configurations, and scaling settings, ensuring a robust foundation for deploying containerized applications with efficiency and security.